

Introduction to Real Analysis

Course Notes

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July 30, 2026

Acknowledgements

The material in these notes is adapted from MATH 321: Real Analysis I, offered by the Department of Mathematics at Carleton College, which I took in Winter 2025 under Professor Kate Meyer. The development of the theory, the choice of examples, and the structure of the proofs follow her course handouts closely, and the figures reproduced here in `TikZ` are redrawings of her hand-drawn diagrams. I am grateful to Professor Meyer for both — and for a course that made the subject feel inevitable rather than arbitrary. Any errors introduced in the transcription are my own.

The conversion of my handwritten iPad notes into `LATEX` — the typesetting, the theorem–proof–example–remark environments, and the redrawing of every diagram in `TikZ` — was carried out with Claude (Anthropic), which is also responsible for the formatting and layout of this document.

Contents

Chapter 1: Sequences, Limits, and the Topology of $\mathbb{R}$	5
1 Completeness of $\mathbb{R}$: How $\mathbb{N}$ and $\mathbb{Q}$ Sit Inside $\mathbb{R}$	5
1.1 The Axiom of Completeness	5
1.2 The Archimedean Property	6
1.3 Density of $\mathbb{Q}$ in $\mathbb{R}$	6
2 Convergence of a Sequence of Real Numbers	7
2.1 Motivating the definition	7
2.2 First consequences	8
3 The Algebraic and Order Limit Theorems	8
3.1 Proof of the sum rule (ii)	8
3.2 Proof of the product rule (iii)	9
3.3 The Order Limit Theorem	9
4 The Monotone Convergence Theorem and an Introduction to Series	10
4.1 Vocabulary for sequences	10
4.2 Infinite series	11
5 Subsequences and the Bolzano–Weierstrass Theorem	12
5.1 Subsequences	12
5.2 Bolzano–Weierstrass	13
6 The Cauchy Criterion and Completeness, Revisited	14
6.1 Two definitions side by side	14
6.2 The completeness picture	15
6.3 Practice with series	16
7 Open and Closed Sets	16
7.1 Open sets	16
7.2 Limit points and closed sets	16
7.3 Building new open and closed sets	17
7.4 The Cantor set	18
7.5 Closure	19
8 Compact Sets	19
8.1 Why compactness	19
8.2 Three equivalent characterizations	19
8.3 Exercises	20
Chapter 2: Functional Limits and Continuity	21

9	Continuous Functions	21
9.1	From functional limits to continuity	21
9.2	The sequential characterization	22
9.3	Building new continuous functions	23
10	Properties of Continuous Functions	24
10.1	Mapping sets forward	24
10.2	Uniform continuity	25
	 Chapter 3: The Derivative	 26
11	The Derivative	26
11.1	Continuity versus differentiability	27
11.2	Derivative rules	27
12	The Mean Value Theorems	29
12.1	Corollaries: inferring f from f'	31
	 Chapter 4: Sequences and Series of Functions	 31
13	Sequences of Functions	31
13.1	Pointwise convergence	32
13.2	Uniform convergence	32
13.3	The Cauchy criterion and the Differentiable Limit Theorem	34
14	Series of Functions	35
15	Power Series	37
16	Taylor Series	39
16.1	Deriving the coefficients	39
16.2	The good, the bad, and the ugly	39
16.3	Remainder theorems	40
	 Chapter 5: The Riemann Integral	 41
17	Definitions: Partitions, Sums, and Integrals	41
18	Which Functions Are Integrable?	43
19	Properties of the Integral	46
19.1	Excusing an endpoint	46
19.2	Algebraic and order properties	47
19.3	Passing limits inside the integral	48

20 The Fundamental Theorem of Calculus	49
Chapter 6: Constructing $\mathbb{R}$ from $\mathbb{Q}$	51
21 Where We Have Been	51
22 Cauchy Sequences of Rationals	51
23 Equivalence Classes of Cauchy Sequences	52
24 Defining $\mathbb{R}$	53
24.1 Confirming the properties of $\mathbb{R}$	53
24.2 Conclusion	54

Chapter 1: Sequences, Limits, and the Topology of $\mathbb{R}$

This chapter develops the basic theory of convergence in $\mathbb{R}$. Everything rests on a single assumption, the *Axiom of Completeness*; the recurring theme is that each major theorem below is, in a precise sense, a repackaging of that one axiom. Section references in the margin notes point to Abbott, *Understanding Analysis*.

1 Completeness of $\mathbb{R}$: How $\mathbb{N}$ and $\mathbb{Q}$ Sit Inside $\mathbb{R}$

1.1 The Axiom of Completeness

Definition 1.1 (Bounds and suprema). Let $A \subseteq \mathbb{R}$. A number $b \in \mathbb{R}$ is an *upper bound* for A if $a \leq b$ for all $a \in A$; A is *bounded above* if such a b exists. A number $s \in \mathbb{R}$ is a *least upper bound* (or *supremum*) of A , written $s = \sup A$, if

1. s is an upper bound for A , and
2. whenever b is an upper bound for A , we have $s \leq b$.

Lower bounds and the *infimum* $\inf A$ are defined symmetrically.

Axiom 1.2 (Axiom of Completeness, AoC). *Every nonempty set of real numbers that is bounded above has a least upper bound in $\mathbb{R}$.*

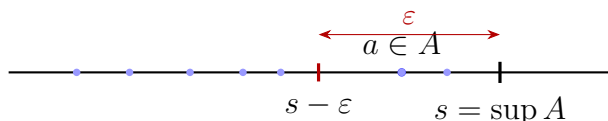
The following reformulation of “least” is the workhorse of nearly every proof in this chapter: it converts the abstract minimality of $\sup A$ into the concrete statement that we may always find an element of A within ε of the supremum.

Lemma 1.3 (Abbott, Lemma 1.3.8). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $s \in \mathbb{R}$ be an upper bound for A . Then*

$$s = \sup A \iff \text{for every } \varepsilon > 0 \text{ there exists } a \in A \text{ with } s - \varepsilon < a.$$

Proof. ($\Rightarrow$) Suppose $s = \sup A$ and let $\varepsilon > 0$. Since $s - \varepsilon < s$ and s is the *least* upper bound, $s - \varepsilon$ is not an upper bound for A . Hence some $a \in A$ satisfies $a > s - \varepsilon$.

($\Leftarrow$) Suppose the ε -condition holds and let b be any upper bound for A . If $b < s$, put $\varepsilon = s - b > 0$; by hypothesis there is $a \in A$ with $a > s - \varepsilon = b$, contradicting that b bounds A . Hence $s \leq b$, and s is the least upper bound. ■



1.2 The Archimedean Property

Theorem 1.4 (Archimedean Property).

- (i) For every $x \in \mathbb{R}$ there exists $n \in \mathbb{N}$ satisfying $n > x$.
- (ii) For every real number $y > 0$ there exists $n \in \mathbb{N}$ satisfying $\frac{1}{n} < y$.

Proof of (i). Assume, for the sake of contradiction, that no such n exists; that is, $n \leq x$ for every $n \in \mathbb{N}$. Then x is an upper bound for $\mathbb{N}$. Since $\mathbb{N}$ is nonempty and bounded above, the Axiom of Completeness (Axiom 1.2) supplies $\alpha = \sup \mathbb{N} \in \mathbb{R}$.

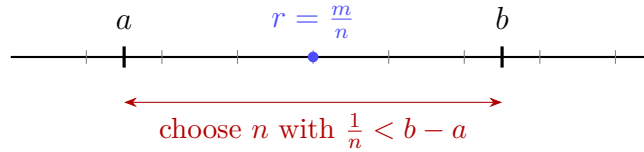
Because α is the *least* upper bound, $\alpha - 1$ is *not* an upper bound for $\mathbb{N}$ (this is Lemma 1.3 with $\varepsilon = 1$). Hence there exists $n \in \mathbb{N}$ with $\alpha - 1 < n$, equivalently $\alpha < n + 1$. But $n + 1 \in \mathbb{N}$, so we have produced an element of $\mathbb{N}$ strictly larger than α , contradicting that α is an upper bound for $\mathbb{N}$. ■

Proof of (ii). Let $y > 0$ and apply part (i) to the real number $x = 1/y$: there is $n \in \mathbb{N}$ with $n > 1/y$. Since $n > 0$ and $y > 0$, multiplying through by $y/n > 0$ gives $y > 1/n$, as desired. ■

Remark 1.5. Part (ii) is the statement that “ $\mathbb{N}$ is not bounded above” in disguise: no matter how small y is, the sequence $1/n$ eventually gets underneath it. It is the reason $(1/n) \rightarrow 0$, and it is used constantly in what follows.

1.3 Density of $\mathbb{Q}$ in $\mathbb{R}$

Theorem 1.6 (Density of $\mathbb{Q}$ in $\mathbb{R}$). For every two real numbers $a < b$ there exists a rational number $r \in \mathbb{Q}$ satisfying $a < r < b$.



Proof. By the Archimedean Property (Theorem 1.4(ii)) choose $n \in \mathbb{N}$ with

$$\frac{1}{n} < b - a.$$

The idea is that a grid of mesh $1/n$ is too fine to step over the interval (a, b) . Applying Theorem 1.4(i) twice, the set $\{k \in \mathbb{Z} : k > na\}$ is nonempty and bounded below, so it has a least element m ; thus

$$m - 1 \leq na < m.$$

The right inequality gives $a < m/n$. From the left inequality, $m \leq na + 1$, and since $1 < n(b - a)$ we get

$$m \leq na + 1 < na + n(b - a) = nb,$$

so $m/n < b$. Therefore $r = m/n \in \mathbb{Q}$ satisfies $a < r < b$. ■

Corollary 1.7 (Density of the irrationals). *The set of irrational numbers $\mathbb{I} = \mathbb{R} \setminus \mathbb{Q}$ is also dense in $\mathbb{R}$: for every $a < b$ there exists $t \in \mathbb{I}$ with $a < t < b$.*

Proof. Apply Theorem 1.6 to the pair $a - \sqrt{2} < b - \sqrt{2}$ to obtain $r \in \mathbb{Q}$ with $a - \sqrt{2} < r < b - \sqrt{2}$. Then $t = r + \sqrt{2}$ lies in (a, b) , and t is irrational, since $t \in \mathbb{Q}$ would force $\sqrt{2} = t - r \in \mathbb{Q}$. ■

2 Convergence of a Sequence of Real Numbers

2.1 Motivating the definition

Definition 2.1 (Sequence). A *sequence of real numbers* is a function $f : \mathbb{N} \rightarrow \mathbb{R}$. We write $a_n = f(n)$ and denote the sequence by $(a_n)_{n=1}^{\infty}$ or simply (a_n) .

We want to make precise the phrase “ (a_n) converges to a ”. Two failed drafts help isolate what the right statement must say.

Draft 1. “*The terms a_n get close to a .*” Too vague: it does not say how close, and it does not say that they *stay* close.

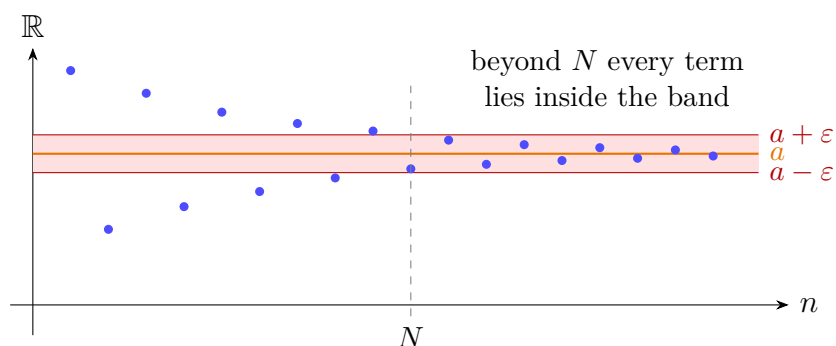
Draft 2. “*Eventually all terms lie within 0.0001 of a .*” Better — it has a tolerance and an “eventually” — but 0.0001 is arbitrary. Convergence should hold for *every* tolerance.

Replacing the fixed tolerance by an arbitrary $\varepsilon > 0$ gives the definition.

Definition 2.2 (Convergence). The sequence $(a_n)_{n=1}^{\infty}$ *converges* to $a \in \mathbb{R}$ if

$$\forall \varepsilon > 0 \quad \exists N \in \mathbb{N} \text{ such that } \forall n \geq N : |a_n - a| < \varepsilon.$$

We then write $\lim_{n \rightarrow \infty} a_n = a$, or $(a_n) \rightarrow a$. A sequence *diverges* if there is no $a \in \mathbb{R}$ to which it converges.



Notation 2.3 (ε -neighborhood). For $x \in \mathbb{R}$ and $\varepsilon > 0$ write

$$V_\varepsilon(x) = \{y \in \mathbb{R} : |y - x| < \varepsilon\} = (x - \varepsilon, x + \varepsilon).$$

Definition 2.2 then reads: for every $\varepsilon > 0$ there is N such that $a_n \in V_\varepsilon(a)$ for all $n \geq N$; equivalently, every ε -neighborhood of a contains all but finitely many terms of the sequence.

2.2 First consequences

Theorem 2.4 (Uniqueness of limits). *If a sequence converges, its limit is unique.*

Proof. Suppose $(a_n) \rightarrow a$ and $(a_n) \rightarrow b$ with $a \neq b$, and set $\varepsilon = |a - b|/2 > 0$. Choose N_a with $|a_n - a| < \varepsilon$ for $n \geq N_a$, and N_b with $|a_n - b| < \varepsilon$ for $n \geq N_b$. For $n \geq \max\{N_a, N_b\}$ the triangle inequality gives

$$|a - b| \leq |a - a_n| + |a_n - b| < \varepsilon + \varepsilon = |a - b|,$$

which is absurd. Hence $a = b$. ■

Example 2.5 (True or false?). “If a sequence contains an infinite number of ones, it must converge to one.” **False.** The sequence

$$(1, 2, 1, 3, 1, 4, 1, 5, \dots)$$

contains infinitely many ones but is unbounded, hence divergent. Containing infinitely many copies of a value says nothing about the *other* terms; Definition 2.2 requires *all* terms past some index to be close to the limit.

3 The Algebraic and Order Limit Theorems

Theorem 3.1 (Algebraic Limit Theorem, ALT). *Suppose $\lim_{n \rightarrow \infty} a_n = a$ and $\lim_{n \rightarrow \infty} b_n = b$. Then*

- (i) $\lim(ca_n) = ca$ for every $c \in \mathbb{R}$;
- (ii) $\lim(a_n + b_n) = a + b$;
- (iii) $\lim(a_n b_n) = ab$;
- (iv) $\lim(a_n/b_n) = a/b$, provided $b \neq 0$.

Remark 3.2. These are the sequence analogues of the limit laws from calculus (Stewart, p. 95). The difference is that here we prove them from Definition 2.2 rather than assert them.

3.1 Proof of the sum rule (ii)

Scratch work. **Want:** $\forall \varepsilon > 0 \exists N$ with $|(a_n + b_n) - (a + b)| < \varepsilon$ for $n \geq N$.

Know: $\forall \varepsilon > 0 \exists N_a$ with $|a_n - a| < \varepsilon$ for $n \geq N_a$; likewise N_b for (b_n) .

Play with the quantity:

$$|(a_n + b_n) - (a + b)| = |(a_n - a) + (b_n - b)| \leq |a_n - a| + |b_n - b|.$$

Make each piece $< \varepsilon/2$, and take $N = \max\{N_a, N_b\}$.

Proof of Theorem 3.1(ii). Let $\varepsilon > 0$. Since $(a_n) \rightarrow a$, choose $N_a \in \mathbb{N}$ so that $|a_n - a| < \varepsilon/2$ whenever $n \geq N_a$; since $(b_n) \rightarrow b$, choose $N_b \in \mathbb{N}$ so that $|b_n - b| < \varepsilon/2$ whenever $n \geq N_b$. Put $N = \max\{N_a, N_b\}$. Then for all $n \geq N$,

$$|(a_n + b_n) - (a + b)| = |(a_n - a) + (b_n - b)| \leq |a_n - a| + |b_n - b| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \quad \blacksquare$$

3.2 Proof of the product rule (iii)

Scratch work. The natural algebraic move is to add and subtract ab_n :

$$|a_nb_n - ab| = |a_nb_n - ab_n + ab_n - ab| = |b_n(a_n - a) + a(b_n - b)| \leq |b_n||a_n - a| + |a||b_n - b|.$$

The factor $|b_n|$ is harmless because a convergent sequence is bounded: there is $M > 0$ with $|b_n| \leq M$ for all n . Then force $|a_n - a| < \varepsilon/(2M)$ and $|b_n - b| < \varepsilon/(2(|a| + 1))$.

Lemma 3.3. *Every convergent sequence is bounded.*

Proof. Let $(b_n) \rightarrow b$ and apply Definition 2.2 with $\varepsilon = 1$: there is N with $|b_n - b| < 1$ for $n \geq N$, hence $|b_n| \leq |b_n - b| + |b| < 1 + |b|$ for such n . Taking

$$M = \max\{|b_1|, |b_2|, \dots, |b_{N-1}|, |b| + 1\}$$

gives $|b_n| \leq M$ for every $n \in \mathbb{N}$. $\blacksquare$

Proof of Theorem 3.1(iii). By Lemma 3.3 pick $M > 0$ with $|b_n| \leq M$ for all n . Let $\varepsilon > 0$. Choose N_a so that

$$|a_n - a| < \frac{\varepsilon}{2M} \quad \text{for } n \geq N_a,$$

and choose N_b so that

$$|b_n - b| < \frac{\varepsilon}{2(|a| + 1)} \quad \text{for } n \geq N_b.$$

Set $N = \max\{N_a, N_b\}$. For $n \geq N$,

$$|a_nb_n - ab| \leq |b_n||a_n - a| + |a||b_n - b| < M \cdot \frac{\varepsilon}{2M} + |a| \cdot \frac{\varepsilon}{2(|a| + 1)} < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \quad \blacksquare$$

3.3 The Order Limit Theorem

Theorem 3.4 (Order Limit Theorem, OLT). *Suppose $\lim a_n = a$ and $\lim b_n = b$.*

- (i) *If $a_n \geq 0$ for all $n \in \mathbb{N}$, then $a \geq 0$.*
- (ii) *If $a_n \leq b_n$ for all $n \in \mathbb{N}$, then $a \leq b$.*
- (iii) *If there exists $c \in \mathbb{R}$ with $c \leq b_n$ for all n , then $c \leq b$; likewise if $a_n \leq c$ for all n , then $a \leq c$.*

Proof. (i) Suppose instead $a < 0$ and take $\varepsilon = |a| = -a > 0$. There is N with $|a_N - a| < \varepsilon$, hence $a_N < a + \varepsilon = 0$, contradicting $a_N \geq 0$.

(ii) The sequence $(b_n - a_n)$ is nonnegative and, by Theorem 3.1, converges to $b - a$. By (i), $b - a \geq 0$, i.e. $a \leq b$.

(iii) Apply (ii) to the constant sequence $a_n = c$ (respectively $b_n = c$), whose limit is c . ■

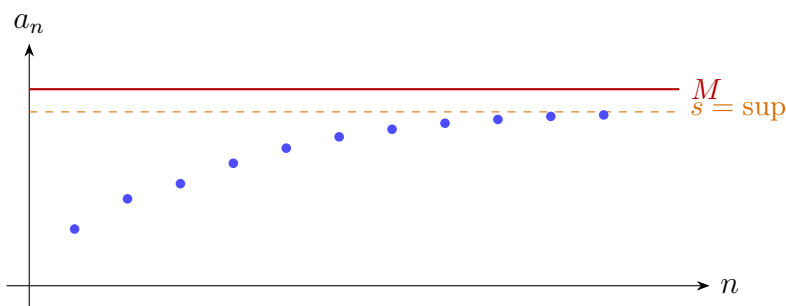
4 The Monotone Convergence Theorem and an Introduction to Series

4.1 Vocabulary for sequences

Definition 4.1. Let (a_n) be a sequence of real numbers.

- (a_n) is *bounded* if there exists $M > 0$ such that $|a_n| \leq M$ (equivalently $-M \leq a_n \leq M$) for all $n \in \mathbb{N}$.
- (a_n) is *increasing* if $a_n \leq a_{n+1}$ for all $n \in \mathbb{N}$.
- (a_n) is *decreasing* if $a_n \geq a_{n+1}$ for all $n \in \mathbb{N}$.
- (a_n) is *monotone* if it is increasing or decreasing.

Theorem 4.2 (Monotone Convergence Theorem, MCT). *If a sequence is monotone and bounded, then it converges.*



Scratch work. We must produce a limit out of thin air. The only mechanism we have for manufacturing real numbers is the Axiom of Completeness, so the candidate limit must be $s = \sup\{a_n : n \in \mathbb{N}\}$.

Proof of Theorem 4.2 (increasing case). Let (a_n) be increasing and bounded, so there is $M > 0$ with $|a_n| < M$ for every n . Consider the set

$$A = \{a_n : n \in \mathbb{N}\}.$$

A is nonempty, and it is bounded above (by M). By the Axiom of Completeness the supremum of A exists; let $s = \sup A$. We claim $(a_n) \rightarrow s$.



Fix $\varepsilon > 0$. By Lemma 1.3, $s - \varepsilon$ is not an upper bound for A , so there exists $N \in \mathbb{N}$ with

$$s - \varepsilon < a_N.$$

Since (a_n) is increasing, whenever $n \geq N$ we have

$$s - \varepsilon < a_N \leq a_n.$$

On the other hand s is an upper bound for A , so for *all* $n \in \mathbb{N}$,

$$a_n \leq s < s + \varepsilon.$$

Combining the two displays, for every $n \geq N$,

$$s - \varepsilon < a_n < s + \varepsilon \iff -\varepsilon < a_n - s < \varepsilon \iff |a_n - s| < \varepsilon,$$

as desired. The decreasing case follows by applying this argument to $(-a_n)$, or by using $\inf$ in place of $\sup$. ■

Remark 4.3. The MCT is powerful precisely because it certifies convergence *without naming the limit*. This is exactly the situation for most series.

4.2 Infinite series

Definition 4.4 (Series and partial sums). Let (b_k) be a sequence. The *infinite series*

$$\sum_{k=1}^{\infty} b_k = b_1 + b_2 + b_3 + \cdots$$

is the formal expression whose associated *sequence of partial sums* is $(s_n)_{n=1}^{\infty}$, where

$$s_n = b_1 + b_2 + \cdots + b_n.$$

Definition 4.5 (Convergence of a series). We say the series $\sum_{k=1}^{\infty} b_k$ *converges to* B if the sequence of partial sums (s_n) converges to B . In that case we write $\sum_{k=1}^{\infty} b_k = B$.

Example 4.6. The series $\sum_{k=1}^{\infty} (-1)^k$ has partial sums

$$(s_n) = (-1, 0, -1, 0, \dots),$$

which diverges by the Divergence Criterion below (Corollary 5.4). Hence the series diverges.

Example 4.7 ($\sum 1/k^2$ converges). The partial sums $s_n = \sum_{k=1}^n 1/k^2$ form an increasing sequence, since each added term $1/k^2$ is positive. For boundedness, compare each term with a telescoping one:

$$s_n = \frac{1}{1 \cdot 1} + \frac{1}{2 \cdot 2} + \frac{1}{3 \cdot 3} + \cdots + \frac{1}{n \cdot n} < \frac{1}{1 \cdot 1} + \frac{1}{2 \cdot 1} + \frac{1}{3 \cdot 2} + \cdots + \frac{1}{n(n-1)}.$$

Since $\frac{1}{n(n-1)} = \frac{1}{n-1} - \frac{1}{n}$, the right-hand side telescopes:

$$1 + \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n-1} - \frac{1}{n}\right) = 2 - \frac{1}{n}.$$

Hence $0 < s_n < 2$ for all n , so (s_n) is bounded. Being monotone and bounded, (s_n) converges by the MCT (Theorem 4.2), and therefore the series $\sum_{k=1}^{\infty} 1/k^2$ converges.

Note that this argument never identifies the sum. (It is $\pi^2/6$, the “Basel problem” solved by Euler; see Abbott §6.1.)

Example 4.8 (The harmonic series diverges). The series $\sum_{k=1}^{\infty} 1/k$ diverges. Grouping terms in blocks of lengths 1, 2, 4, 8, $\dots$,

$$1 + \frac{1}{2} + \underbrace{\frac{1}{3} + \frac{1}{4}}_{> 1/2} + \underbrace{\frac{1}{5} + \cdots + \frac{1}{8}}_{> 1/2} + \cdots,$$

each block contributes more than $1/2$, so the partial sums are unbounded. Compare Abbott, Example 2.4.5.

5 Subsequences and the Bolzano–Weierstrass Theorem

5.1 Subsequences

Definition 5.1 (Subsequence). Let $(a_n)_{n=1}^{\infty} = (a_1, a_2, a_3, \dots)$ be a sequence of real numbers and let $n_1 < n_2 < n_3 < \cdots$ be a *strictly increasing* sequence of natural numbers. The sequence

$$(a_{n_k})_{k=1}^{\infty} = (a_{n_1}, a_{n_2}, a_{n_3}, \dots)$$

is called a *subsequence* of $(a_n)_{n=1}^{\infty}$.

Example 5.2. Consider

$$(a_n)_{n=1}^{\infty} = \left(\frac{(-1)^{n+1}}{n} \right)_{n=1}^{\infty} = \left(1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \frac{1}{5}, -\frac{1}{6}, \dots \right).$$

- $\left(1, \frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \dots \right)$ is a subsequence, with $n_k = 2k - 1$:

k	1	2	3	4
n_k	1	3	5	7

- $(1, -\frac{1}{2}, \frac{1}{3}, \frac{1}{5}, -\frac{1}{8}, \dots)$ **is** a subsequence, with

k	1	2	3	4	5
n_k	1	2	3	5	8

- $(-\frac{1}{2}, 1, -\frac{1}{4}, \frac{1}{3}, \dots)$ is **not** a subsequence: the indices $2, 1, 4, 3, \dots$ are not increasing. Order may not be rearranged.
- $(1, 1, -\frac{1}{2}, \frac{1}{3}, \dots)$ is **not** a subsequence: it repeats the index $n = 1$, and $n_1 < n_2$ is required to be *strict*.

Theorem 5.3 (Abbott, Thm. 2.5.2). *If $(a_n)_{n=1}^{\infty}$ converges to a , then every subsequence $(a_{n_k})_{k=1}^{\infty}$ converges to the same limit a .*

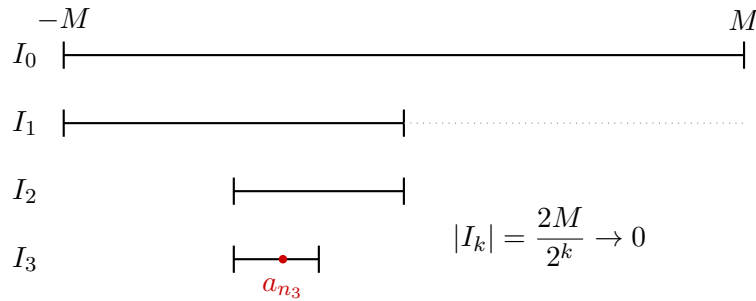
Proof. Let $\varepsilon > 0$ and choose N with $|a_n - a| < \varepsilon$ for all $n \geq N$. Since (n_k) is strictly increasing, an easy induction gives $n_k \geq k$; hence $k \geq N$ implies $n_k \geq N$, so $|a_{n_k} - a| < \varepsilon$. ■

Corollary 5.4 (Divergence Criterion). *The contrapositive of Theorem 5.3 says: if some subsequence of (a_n) fails to converge to a , then (a_n) does not converge to a . In particular, if two subsequences of (a_n) converge to different limits, then (a_n) diverges.*

Example 5.5. $((-1)^n)_{n=1}^{\infty} = (-1, 1, -1, 1, \dots)$ contains the subsequence $(1, 1, 1, \dots) \rightarrow 1$ and the subsequence $(-1, -1, -1, \dots) \rightarrow -1$. Since $1 \neq -1$, the sequence diverges by Corollary 5.4.

5.2 Bolzano–Weierstrass

Theorem 5.6 (Bolzano–Weierstrass). *Every bounded sequence contains a convergent subsequence.*



Proof. Let (a_n) be bounded; by definition there exists $M > 0$ such that $|a_n| \leq M$ for all $n \in \mathbb{N}$. Starting from $I_0 = [-M, M]$, run the following recursive construction, which produces simultaneously a subsequence $(a_{n_k})_{k=1}^{\infty}$ and a nested sequence of closed intervals $I_1 \supseteq I_2 \supseteq I_3 \supseteq \dots$. Take $n_0 = 0$.

1. *Bisect* the current interval I_k into two closed intervals meeting at the midpoint of I_k .

2. Let I_{k+1} be a half of I_k containing infinitely many terms of the sequence. (At least one half must, since I_k contains infinitely many.)
3. Let $a_{n_{k+1}}$ be a term of the sequence lying in I_{k+1} with $n_{k+1} > n_k$. (Possible because I_{k+1} contains infinitely many terms.)

By the Nested Interval Property, the intersection $\bigcap_{k=1}^{\infty} I_k$ contains at least one element; call it x . We claim $(a_{n_k}) \rightarrow x$.

Fix $\varepsilon > 0$. The length of I_k is $2M/2^k$, and by the Algebraic Limit Theorem together with $\lim_{k \rightarrow \infty} (1/2)^k = 0$,

$$\lim_{k \rightarrow \infty} \frac{2M}{2^k} = 2M \lim_{k \rightarrow \infty} \left(\frac{1}{2}\right)^k = 2M \cdot 0 = 0.$$

Hence there exists $N \in \mathbb{N}$ such that $k \geq N$ implies $|2M/2^k - 0| < \varepsilon$. Now whenever $k \geq N$ we have both $a_{n_k} \in I_k$ and $x \in I_k$, and I_k has length less than ε , so

$$|a_{n_k} - x| < \varepsilon. \quad \blacksquare$$

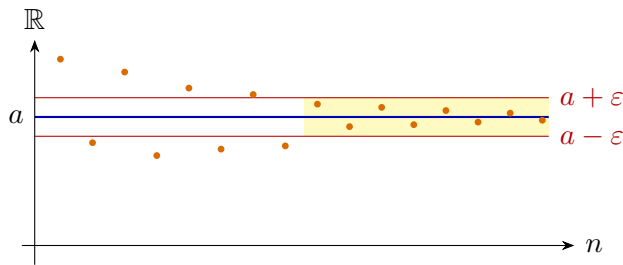
6 The Cauchy Criterion and Completeness, Revisited

6.1 Two definitions side by side

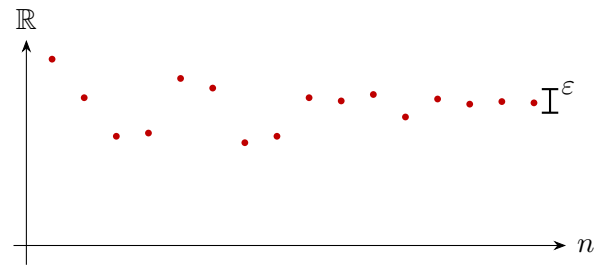
Definition 6.1 (Cauchy sequence). A sequence (a_n) is *Cauchy* if

$$\forall \varepsilon > 0 \quad \exists N \in \mathbb{N} \text{ such that } \forall n, m \geq N : |a_n - a_m| < \varepsilon.$$

Compare with Definition 2.2. The definition of convergence *relies on naming the limit* a ; the Cauchy condition does not mention a limit at all — it only says the terms bunch together. That is what makes it usable for series, whose sums are usually unknown.



convergent:
definition names the limit a



Cauchy:
no limit is named

Theorem 6.2 (Cauchy Criterion for Convergence). *A sequence of real numbers converges if and only if it is Cauchy.*

Proof of $(\Rightarrow)$. Suppose $(a_n) \rightarrow a$ and let $\varepsilon > 0$. Choose N with $|a_n - a| < \varepsilon/2$ for all $n \geq N$. Then for $n, m \geq N$,

$$|a_n - a_m| \leq |a_n - a| + |a - a_m| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \quad \blacksquare$$

Proof of ($\Leftarrow$). Assume (a_n) is Cauchy; we show it converges to some $a \in \mathbb{R}$. The argument has three steps.

Step 1: (a_n) is bounded. Apply Definition 6.1 with $\varepsilon = 1$: choose $N \in \mathbb{N}$ such that $|a_n - a_m| < 1$ whenever $n, m \geq N$. Taking $m = N$ gives $|a_n - a_N| < 1$ for all $n \geq N$, hence

$$|a_n| = |a_n - a_N + a_N| \leq |a_n - a_N| + |a_N| < |a_N| + 1.$$

Letting

$$M = \max\{|a_1|, |a_2|, \dots, |a_{N-1}|, |a_N| + 1\}$$

we get $|a_n| \leq M$ for all $n \in \mathbb{N}$, so (a_n) is bounded.

Step 2: extract a convergent subsequence. Since (a_n) is bounded, the Bolzano–Weierstrass Theorem (Theorem 5.6) provides a subsequence (a_{n_k}) converging to some real number a .

Step 3: the whole sequence converges to a . Fix $\varepsilon > 0$. By the triangle inequality, for any $n, k \in \mathbb{N}$,

$$|a_n - a| = |a_n - a_{n_k} + a_{n_k} - a| \leq \underbrace{|a_n - a_{n_k}|}_{(I)} + \underbrace{|a_{n_k} - a|}_{(II)}.$$

(I) Since (a_n) is Cauchy, there exists $N \in \mathbb{N}$ with $|a_n - a_{n_k}| < \varepsilon/2$ whenever $n, n_k \geq N$.

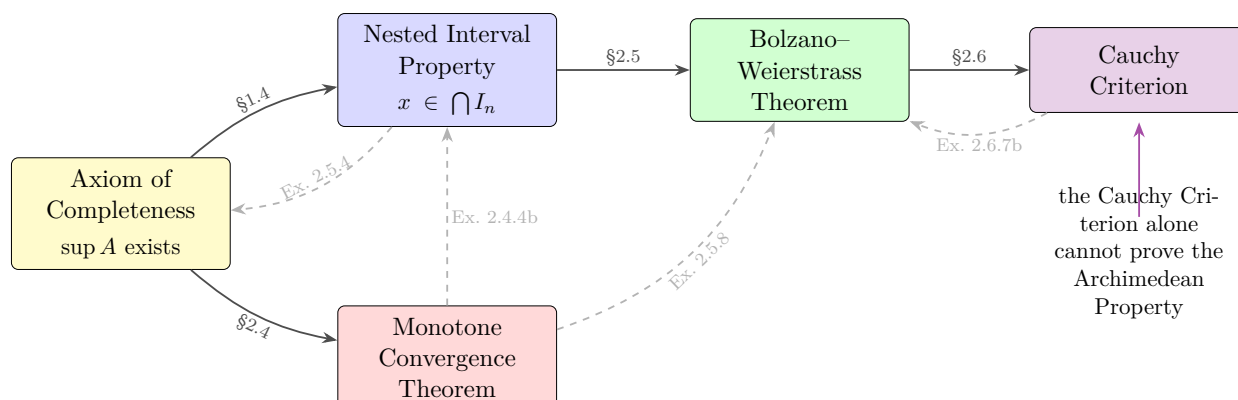
(II) Since $(a_{n_k}) \rightarrow a$, there exists $K \in \mathbb{N}$ with $|a_{n_k} - a| < \varepsilon/2$ whenever $k \geq K$.

Choose one subsequence index n_j with $j \geq K$ and $n_j \geq N$ (possible since $n_k \geq k$). Then for every $n \geq N$ both (I) and (II) are less than $\varepsilon/2$ with this fixed j , so

$$|a_n - a| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \quad \blacksquare$$

6.2 The completeness picture

Every theorem in this chapter is a face of the same axiom. The following map records the logical dependencies as they were developed; solid arrows are the implications proved in class or in Abbott, dashed arrows are the converse directions left as exercises.



6.3 Practice with series

Example 6.3 (True or false?).

1. If $(a_n) \rightarrow 0$ then $\sum_{k=1}^{\infty} a_k$ converges.

False. $(1/n) \rightarrow 0$, but the harmonic series $\sum_{n=1}^{\infty} 1/n$ diverges (Example 4.8).

2. $\sum_{k=1}^{\infty} \frac{k-1}{k}$ diverges.

True. The terms satisfy $\frac{k-1}{k} \rightarrow 1 \neq 0$, so the partial sums cannot converge.

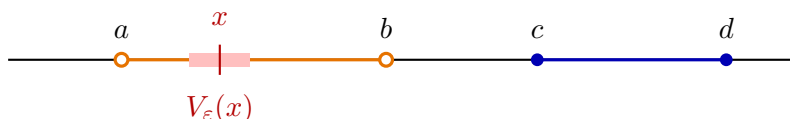
3. $\sum_{k=1}^{\infty} \frac{1+\sqrt{k}}{k}$ converges.

False. For all $k \in \mathbb{N}$, $0 < \frac{1}{k} < \frac{1+\sqrt{k}}{k}$, and $\sum 1/k$ diverges, so the series diverges by comparison.

7 Open and Closed Sets

7.1 Open sets

Definition 7.1 (Open set). A set $A \subseteq \mathbb{R}$ is *open* if for every $x \in A$ there exists $\varepsilon > 0$ such that $V_{\varepsilon}(x) \subseteq A$, where $V_{\varepsilon}(x) = (x - \varepsilon, x + \varepsilon)$ as in Notation 2.3.



Example 7.2. $(a, b) = \{x \in \mathbb{R} : a < x < b\}$ is open: given $x \in (a, b)$, take $\varepsilon = \min\{b - x, x - a\} > 0$; then $V_{\varepsilon}(x) \subseteq (a, b)$.

Remark 7.3. Why do we care about the open/closed distinction? Because $\sup(a, b) = b \notin (a, b)$, whereas $\sup[c, d] = d \in [c, d]$. Suprema of open sets can escape the set. This is also why the Nested Interval Property requires *closed* intervals: $\bigcap_{n=1}^{\infty} (0, \frac{1}{n}) = \emptyset$.

7.2 Limit points and closed sets

Definition 7.4 (Limit point). Given $A \subseteq \mathbb{R}$, a point $x \in \mathbb{R}$ is a *limit point* (or *accumulation point*) of A if for every $\varepsilon > 0$ the neighborhood $V_{\varepsilon}(x)$ contains some element $a \in A$ with $a \neq x$.

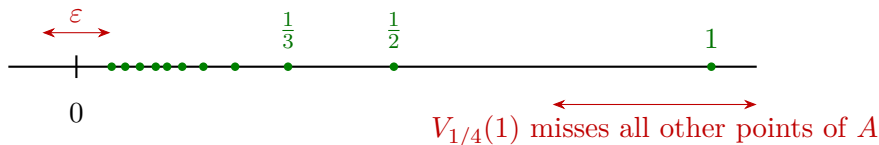
Definition 7.5 (Isolated point). A point $a \in A$ is an *isolated point* of A if it is not a limit point of A .

Definition 7.6 (Closed set). A set $A \subseteq \mathbb{R}$ is *closed* if it contains all of its limit points.

Example 7.7. Let $A = \{\frac{1}{n} : n \in \mathbb{N}\}$.

- Is 1 a limit point of A ? **No.** Taking $\varepsilon = 1/4$, the neighborhood $V_{1/4}(1) = (3/4, 5/4)$ contains no element of A other than 1 itself.

- Is 0 a limit point of A ? **Yes.** Let $\varepsilon > 0$ be arbitrary, so $V_\varepsilon(0) = (-\varepsilon, \varepsilon)$. By the Archimedean Property (Theorem 1.4(ii)) there is $n \in \mathbb{N}$ with $1/n < \varepsilon$, and then $1/n \in V_\varepsilon(0) \cap A$ with $1/n \neq 0$.
- Is A closed? **No**, since 0 is a limit point of A but $0 \notin A$.



Example 7.8. Is $(0, 1]$ open? Closed? **Neither.**

- Not closed: 0 is a limit point of $(0, 1]$ but $0 \notin (0, 1]$.
- Not open: $1 \in (0, 1]$, but every neighborhood $V_\varepsilon(1)$ contains points > 1 , so $V_\varepsilon(1) \not\subseteq (0, 1]$.

Remark 7.9. Sets in $\mathbb{R}$ can be open, closed, neither, or *both*. (What are the two subsets of $\mathbb{R}$ that are both?)

The next theorem explains the word “closed”: closed sets are exactly the sets that are closed under taking limits of sequences.

Theorem 7.10 (Sequential characterization of limit points). *A point $x \in \mathbb{R}$ is a limit point of $A \subseteq \mathbb{R}$ if and only if $x = \lim_{n \rightarrow \infty} a_n$ for some sequence (a_n) of elements of A satisfying $a_n \neq x$ for all n .*

Proof sketch. ($\Leftarrow$) If $a_n \rightarrow x$ with $a_n \in A \setminus \{x\}$, then every $V_\varepsilon(x)$ contains some $a_n \neq x$. ($\Rightarrow$) If x is a limit point, then for each $n \in \mathbb{N}$ choose $a_n \in V_{1/n}(x) \cap A$ with $a_n \neq x$; then $|a_n - x| < 1/n \rightarrow 0$. Full details: Abbott, pp. 89–90. ■

Example 7.11. The interval $[c, d]$ is closed. Indeed, suppose $x = \lim a_n$ with $a_n \in [c, d]$ and $a_n \neq x$. From $c \leq a_n \leq d$ and the Order Limit Theorem (Theorem 3.4),

$$\lim_{n \rightarrow \infty} c \leq \lim_{n \rightarrow \infty} a_n \leq \lim_{n \rightarrow \infty} d \implies c \leq x \leq d,$$

so $x \in [c, d]$.

Example 7.12 (Limit points of $\mathbb{Q}$). *Every real number is a limit point of $\mathbb{Q}$: given $x \in \mathbb{R}$ and $\varepsilon > 0$, density of $\mathbb{Q}$ (Theorem 1.6) supplies a rational q with $x - \varepsilon < q < x + \varepsilon$.*

- $\mathbb{Q}$ is **not closed**: its limit points make up all of $\mathbb{R}$, and $\mathbb{Q} \neq \mathbb{R}$.
- $\mathbb{Q}$ is **not open**: given $q \in \mathbb{Q}$ and $\varepsilon > 0$, density of the irrationals (Corollary 1.7) supplies an irrational j with $q - \varepsilon < j < q + \varepsilon$, so $V_\varepsilon(q) \not\subseteq \mathbb{Q}$.

7.3 Building new open and closed sets

Theorem 7.13 (Abbott, Thm. 3.2.3).

- (i) *The union of an arbitrary collection of open sets is open. Dually, the union of a finite collection of closed sets is closed.*
- (ii) *The intersection of a finite collection of open sets is open. Dually, the intersection of an arbitrary collection of closed sets is closed.*

Proof. (i) If $x \in \bigcup_{\lambda} O_{\lambda}$ then $x \in O_{\lambda_0}$ for some λ_0 , and an ε that works for O_{λ_0} works for the union.

(ii) If $x \in O_1 \cap \cdots \cap O_n$, choose $\varepsilon_i > 0$ with $V_{\varepsilon_i}(x) \subseteq O_i$ and set $\varepsilon = \min\{\varepsilon_1, \dots, \varepsilon_n\} > 0$. The minimum of finitely many positive numbers is positive — this is exactly where finiteness is needed.

The dual statements follow from Theorem 7.15 together with De Morgan's laws. ■

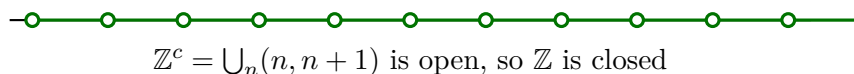
Exercise 7.14. Find an infinite intersection of open sets that is not open. *Answer:* $\bigcap_{n=1}^{\infty} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\}$.

Theorem 7.15 (Complements). *A set $B \subseteq \mathbb{R}$ is open if and only if B^c is closed; a set $C \subseteq \mathbb{R}$ is closed if and only if C^c is open.*

Example 7.16. $\mathbb{Z}$ is closed, because

$$\mathbb{Z}^c = \bigcup_{n=-\infty}^{\infty} (n, n+1)$$

is a union of open intervals, hence open by Theorem 7.13(i).



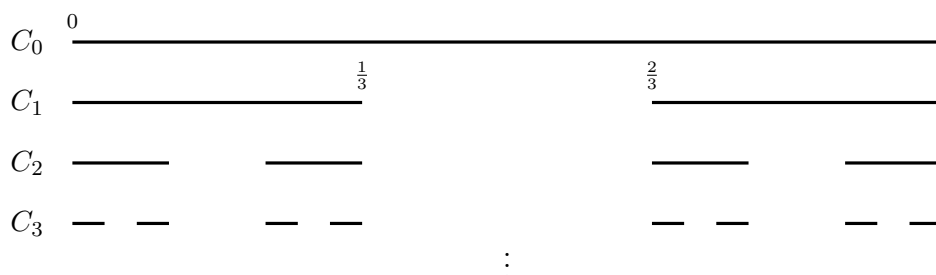
7.4 The Cantor set

Definition 7.17 (Cantor set). Set $C_0 = [0, 1]$ and, recursively,

$$C_{n+1} = C_n \setminus (\text{the open middle thirds of the intervals of } C_n),$$

so that $C_1 = [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$, and so on. The *Cantor set* is

$$\mathcal{C} = \bigcap_{n=0}^{\infty} C_n.$$



Proposition 7.18. *The Cantor set $\mathcal{C}$ is closed.*

Proof. Each C_n is a finite union of closed intervals, hence closed (Theorem 7.13(i)); equivalently, C_n is the complement of a union of open intervals. An arbitrary intersection of closed sets is closed by Theorem 7.13(ii), so $\mathcal{C} = \bigcap_n C_n$ is closed. ■

7.5 Closure

Definition 7.19 (Closure). The *closure* of $A \subseteq \mathbb{R}$ is the union of A with its set of limit points. It is denoted $\overline{A}$ or $\text{Cl}(A)$.

Theorem 7.20. *For any $A \subseteq \mathbb{R}$, the closure $\overline{A}$ is closed, and it is the smallest closed set containing A : if F is closed and $A \subseteq F$, then $\overline{A} \subseteq F$.*

Remark 7.21. By Example 7.12, $\overline{\mathbb{Q}} = \mathbb{R}$. This is the precise sense in which “ $\mathbb{Q}$ is dense in $\mathbb{R}$ ”: a subset is dense exactly when its closure is the whole space.

8 Compact Sets

8.1 Why compactness

Many theorems stated for a closed bounded interval $[a, b]$ are really theorems about compact sets. Two examples:

- **Nested Interval Property $\rightsquigarrow$ Nested Compact Set Property:** if $K_1 \supseteq K_2 \supseteq K_3 \supseteq \cdots$ is a nested sequence of nonempty compact subsets of $\mathbb{R}$, then $\bigcap_{n=1}^{\infty} K_n$ is not empty.
- **Extreme Value Theorem:** if f is continuous on a closed interval $[a, b]$, it attains an absolute maximum and an absolute minimum on $[a, b]$. This remains true with $[a, b]$ replaced by any compact set.

Definition 8.1 (Bounded set). A set $A \subseteq \mathbb{R}$ is *bounded* if there exists $M > 0$ such that $|a| \leq M$ for all $a \in A$.

8.2 Three equivalent characterizations

Definition 8.2 (Compactness, sequential form). A set $K \subseteq \mathbb{R}$ is *compact* if every sequence in K has a subsequence that converges to a limit *that is also in* K .

Theorem 8.3 (Abbott, Thm. 3.3.4, and Heine–Borel). *For $K \subseteq \mathbb{R}$ the following are equivalent:*

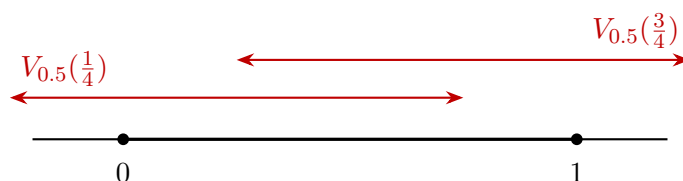
- (a) K is compact in the sense of Definition 8.2;
- (b) K is closed and bounded;
- (c) every open cover of K admits a finite subcover.

Definition 8.4 (Open cover, finite subcover). An *open cover* of $A \subseteq \mathbb{R}$ is a collection $\mathcal{O} = \{O_\lambda\}_{\lambda \in \Lambda}$ of open sets whose union contains A , that is $\bigcup_{\lambda \in \Lambda} O_\lambda \supseteq A$. The indexing set Λ may be finite, countably infinite (e.g. $\Lambda = \mathbb{N}$), or uncountable (e.g. $\Lambda = [0, 1]$).

Given an open cover $\{O_\lambda\}_{\lambda \in \Lambda}$ of A , a *finite subcover* is a finite subcollection of the original cover whose union still contains A .

Example 8.5 (Covers of $[0, 1]$).

- $\mathcal{O} = \{(-\frac{1}{4}, \frac{3}{4}), (\frac{1}{4}, \frac{5}{4})\}$ is a finite open cover of $[0, 1]$.
- $\mathcal{O} = \{V_{0.5}(x)\}_{x \in [0, 1]}$ is an uncountable open cover. A finite subcover is $\{V_{0.5}(\frac{1}{4}), V_{0.5}(\frac{3}{4})\}$, since $(-\frac{1}{4}, \frac{3}{4}) \cup (\frac{1}{4}, \frac{5}{4}) \supseteq [0, 1]$.

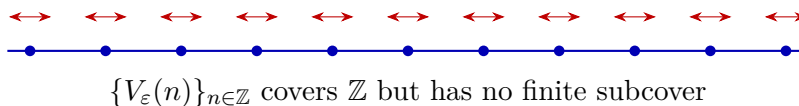


Example 8.6 ($\mathbb{Z}$ is not compact). Since $\mathbb{Z}$ is unbounded it cannot be compact, so by Theorem 8.3(c) there must exist an open cover of $\mathbb{Z}$ with no finite subcover. One is

$$\mathcal{O} = \{V_\varepsilon(n)\}_{n \in \mathbb{Z}} \quad \text{for any fixed } 0 < \varepsilon < \frac{1}{2},$$

since each $V_\varepsilon(n)$ contains exactly one integer, so no finite subcollection can cover all of $\mathbb{Z}$.

By contrast, $\mathcal{O} = \{\mathbb{R}\}$ is an open cover of $\mathbb{Z}$, and it trivially has the finite subcover $\{\mathbb{R}\}$. Compactness requires that *every* open cover have a finite subcover, not merely that some cover does.



8.3 Exercises

Exercise 8.7 (Abbott 3.3.2). Decide which of the following sets are compact. For those that are not, exhibit a sequence in the set possessing no subsequence converging to a limit in the set.

- $\mathbb{N}$
- $\mathbb{Q} \cap [0, 1]$
- The Cantor set $\mathcal{C}$
- $\{1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{n^2} : n \in \mathbb{N}\}$
- $\{1, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots\}$

Exercise 8.8 (Abbott 3.3.4). Assume K is compact and F is closed. Decide whether each of the following sets is definitely compact, definitely closed, both, or neither.

- (a) $K \cap F$
- (b) $\overline{F^c \cup K^c}$
- (c) $K \setminus F = \{x \in K : x \notin F\}$
- (d) $\overline{K \cap F^c}$

Chapter 2: Functional Limits and Continuity

Chapter 1 studied limits of *sequences*. We now transplant the same ε - δ machinery to *functions*, and discover that continuity is exactly the property that makes limits and function evaluation commute. The payoff arrives in section 10, where the topological work of section 8 lets us prove the Extreme Value Theorem.

9 Continuous Functions

9.1 From functional limits to continuity

Definition 9.1 (Functional limit). Let $f : A \rightarrow \mathbb{R}$ and let c be a *limit point* of the domain A (Definition 7.4). We write $\lim_{x \rightarrow c} f(x) = L$ to mean

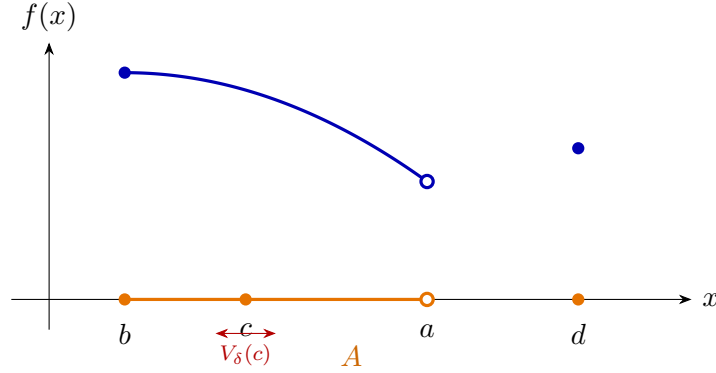
$$\forall \varepsilon > 0 \quad \exists \delta > 0 \quad \text{such that} \quad 0 < |x - c| < \delta \implies |f(x) - L| < \varepsilon,$$

where x ranges over A . Equivalently: $x \in V_\delta(c) \setminus \{c\}$ implies $f(x) \in V_\varepsilon(L)$.

Remark 9.2. Two features are worth pausing on.

- c need *not* belong to A ; it only has to be a limit point, so that there are points of A arbitrarily near c to test.
- The condition $0 < |x - c|$ excludes $x = c$. Whatever value f takes *at* c — if any — does not affect $\lim_{x \rightarrow c} f(x)$.

Example 9.3. Consider $f : A \rightarrow \mathbb{R}$ pictured below, with domain $A = [b, a) \cup \{d\}$.



- $\lim_{x \rightarrow a} f(x)$ **exists**: a is a limit point of A , and it is fine that $a \notin A$.
- $\lim_{x \rightarrow b} f(x)$ **exists**, taking one-sided approach inside A .
- $\lim_{x \rightarrow d} f(x)$ **does not exist**: d is an isolated point of A , not a limit point, so Definition 9.1 does not apply.

We now translate the calculus slogan “ f is continuous at c means $\lim_{x \rightarrow c} f(x) = f(c)$ ” into ε - δ form. A first draft would read: for all $\varepsilon > 0$ there is $\delta > 0$ such that $0 < |x - c| < \delta$ implies $|f(x) - f(c)| < \varepsilon$. But the restriction $x \neq c$ is now unnecessary — indeed harmful. We *want* $c \in A$ so that $f(c)$ is defined, and if $x = c$ then $|f(x) - f(c)| = 0 < \varepsilon$ automatically. Dropping it gives the definition and, as a bonus, makes functions continuous at isolated points.

Definition 9.4 (Continuity). A function $f : A \rightarrow \mathbb{R}$ is *continuous at* $c \in A$ if

$$\forall \varepsilon > 0 \quad \exists \delta > 0 \quad \text{such that} \quad |x - c| < \delta \implies |f(x) - f(c)| < \varepsilon$$

for all $x \in A$. We say f is *continuous on* $B \subseteq A$ if f is continuous at every $b \in B$.

Remark 9.5. Unlike Definition 9.1, here c must lie in A but need not be a limit point of A : *every function is continuous at every isolated point of its domain*. In Example 9.3, f is continuous exactly on $[b, a) \cup \{d\}$ — including at the isolated point d .

Example 9.6 ($f(x) = x^2$ is continuous at $x = 3$).

Scratch work. We must control $|x^2 - 9| = |x - 3| |x + 3|$. The first factor is what δ controls; the second must be bounded. Insisting $|x - 3| < 1$ gives $2 < x < 4$, hence $5 < x + 3 < 7$, so $|x + 3| \leq 7$. Then it suffices to also force $|x - 3| < \varepsilon/7$.

Proof. Let $\varepsilon > 0$ be arbitrary and set $\delta = \min\{\varepsilon/7, 1\}$. If $|x - 3| < \delta$ then both $|x - 3| < \varepsilon/7$ and $|x + 3| \leq 7$ hold, so

$$|x^2 - 9| = |x - 3| |x + 3| < \frac{\varepsilon}{7} \cdot 7 = \varepsilon. \quad \blacksquare$$

9.2 The sequential characterization

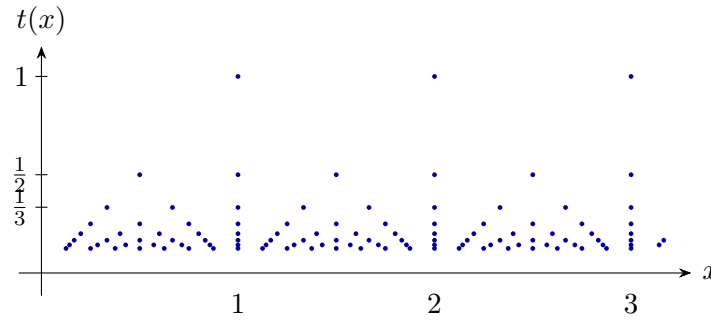
Theorem 9.7 (Sequential Characterization of Continuity). *Let $f : A \rightarrow \mathbb{R}$ and $c \in A$. Then f is continuous at c if and only if for every sequence (x_n) of points of A with $(x_n) \rightarrow c$, the*

image sequence satisfies $(f(x_n))_{n=1}^{\infty} \rightarrow f(c)$.

Corollary 9.8 (Criterion for Discontinuity). *Let $f : A \rightarrow \mathbb{R}$ and let $c \in A$ also be a limit point of A . If there exists a sequence (x_n) in A with $(x_n) \rightarrow c$ but $(f(x_n)) \not\rightarrow f(c)$, then f is not continuous at c .*

Example 9.9 (Thomae's function). Define $t : \mathbb{R} \rightarrow \mathbb{R}$ by

$$t(x) = \begin{cases} 1 & \text{if } x = 0, \\ \frac{1}{n} & \text{if } x \in \mathbb{Q} \setminus \{0\}, x = \frac{m}{n} \text{ in lowest terms with } n > 0, \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$$



Is t continuous at $x = \frac{1}{2}$? **No.** Take the sequence of irrationals

$$x_n = \frac{1}{2} - \frac{\sqrt{2}}{n} \rightarrow \frac{1}{2},$$

for which $t(x_n) = 0$ for every n . Then $(t(x_n)) = (0, 0, 0, \dots) \rightarrow 0 \neq \frac{1}{2} = t(\frac{1}{2})$, so Corollary 9.8 applies. The same argument works at every rational point: t is discontinuous at every rational number.

Is t continuous at $x = \frac{\sqrt{2}}{2}$? **Yes** — and more generally at every irrational. Since $t(\frac{\sqrt{2}}{2}) = 0$, given $\varepsilon > 0$ we must find δ making $|t(x)| < \varepsilon$ nearby. The key observation is that only finitely many points of $[0, 1]$ satisfy $t(x) \geq \varepsilon$: these are the rationals m/n with $1/n \geq \varepsilon$, i.e. $n \leq 1/\varepsilon$, and for each such n there are at most $n+1$ candidates in $[0, 1]$. Choosing δ smaller than the distance from $\frac{\sqrt{2}}{2}$ to this finite set does the job.

9.3 Building new continuous functions

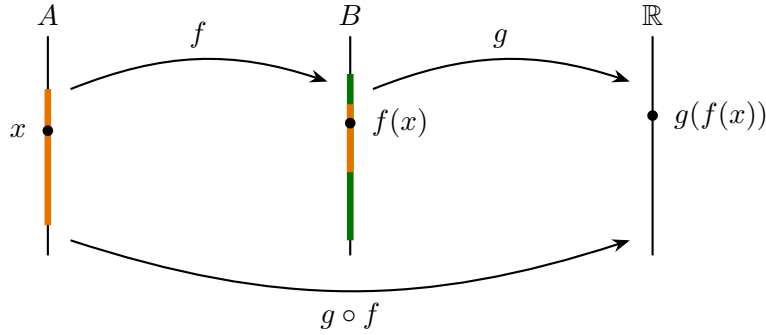
Theorem 9.10 (Algebraic Continuity Theorem, Abbott 4.3.4). *Assume $f : A \rightarrow \mathbb{R}$ and $g : A \rightarrow \mathbb{R}$ are continuous at $c \in A$. Then*

- (i) kf is continuous at c for every $k \in \mathbb{R}$;
- (ii) $f + g$ is continuous at c ;
- (iii) fg is continuous at c ;

(iv) f/g is continuous at c , provided the quotient is defined.

Proof idea. Combine Theorem 9.7 with the Algebraic Limit Theorem (Theorem 3.1): if $x_n \rightarrow c$ then $f(x_n) \rightarrow f(c)$ and $g(x_n) \rightarrow g(c)$, so $f(x_n) + g(x_n) \rightarrow f(c) + g(c)$, and likewise for the other operations. ■

Theorem 9.11 (Composition of Continuous Functions, Abbott 4.3.9). *Given $f : A \rightarrow \mathbb{R}$ and $g : B \rightarrow \mathbb{R}$, assume the range $f(A) = \{f(x) : x \in A\}$ is contained in B , so that $g \circ f(x) = g(f(x))$ is defined on A . If f is continuous at $c \in A$ and g is continuous at $f(c) \in B$, then $g \circ f$ is continuous at c .*



10 Properties of Continuous Functions

10.1 Mapping sets forward

Definition 10.1 (Image of a set). If $f : A \rightarrow \mathbb{R}$ and $B \subseteq A$, the *image* of B under f is

$$f(B) = \{f(b) : b \in B\}.$$

Theorem 10.2 (Preservation of Compact Sets). *Let $f : A \rightarrow \mathbb{R}$ be continuous on $A \subseteq \mathbb{R}$. If $K \subseteq A$ is compact, then $f(K)$ is compact.*

Proof. Let $(y_n)_{n=1}^{\infty}$ be a sequence in $f(K)$; we must produce a subsequence converging to a point of $f(K)$. For each $n \in \mathbb{N}$, membership $y_n \in f(K)$ means $y_n = f(x_n)$ for some $x_n \in K$. Consider the resulting sequence $(x_n)_{n=1}^{\infty}$ in K .

Because K is compact (Definition 8.2), there exists a subsequence $(x_{n_k})_{k=1}^{\infty}$ converging to some $x \in K$. And since f is continuous, Theorem 9.7 gives

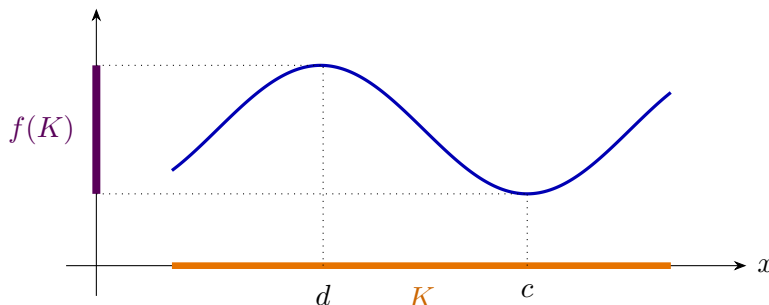
$$(f(x_{n_k}))_{k=1}^{\infty} \longrightarrow f(x).$$

Thus $(f(x_{n_k}))$ is a subsequence of (y_n) converging to $f(x) \in f(K)$, as required. ■

Theorem 10.3 (Extreme Value Theorem). *If $f : K \rightarrow \mathbb{R}$ is continuous on a compact set $K \neq \emptyset$, then there exist $c, d \in K$ such that*

$$f(c) \leq f(x) \leq f(d) \quad \text{for all } x \in K.$$

That is, f attains an absolute minimum value $f(c)$ and an absolute maximum value $f(d)$ on K .



Proof sketch. K compact and f continuous give $f(K)$ compact (Theorem 10.2), hence closed and bounded (Theorem 8.3). Boundedness gives $\sup f(K) \in \mathbb{R}$; closedness gives $\sup f(K) \in \overline{f(K)} = f(K)$, so $\sup f(K) = f(d)$ for some $d \in K$. The minimum is handled symmetrically with $\inf$. ■

10.2 Uniform continuity

Definition 10.4 (Uniform continuity). A function $f : A \rightarrow \mathbb{R}$ is *uniformly continuous* on A if

$$\forall \varepsilon > 0 \quad \exists \delta > 0 \quad \text{such that} \quad \forall x, y \in A : \quad |x - y| < \delta \implies |f(x) - f(y)| < \varepsilon.$$

Compare with Definition 9.4, which says: for all $c \in A$, for all $\varepsilon > 0$, there is $\delta > 0$ such that $|x - c| < \delta$ implies $|f(x) - f(c)| < \varepsilon$. The difference is the order of the quantifiers. In ordinary continuity, δ is chosen *after* c and may shrink as c moves; in uniform continuity a single δ must *work for all points of the domain at once*.

Example 10.5 ($f(x) = x^2$ is not uniformly continuous on $\mathbb{R}$). Fix $\varepsilon = 1$ and ask how large a δ survives at various centres c :

c	largest δ with $ x - c < \delta \implies x^2 - c^2 < 1$
1.5	≈ 0.30
3	≈ 0.16
5	≈ 0.09

The required δ shrinks to 0 as $c \rightarrow \infty$, so no single δ works for all $c \in \mathbb{R}$.

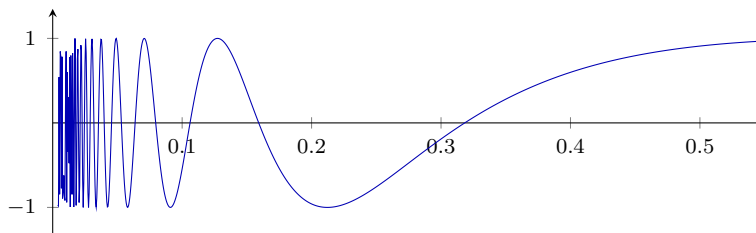
Proposition 10.6. $f(x) = x^2$ is *uniformly continuous* on $[0, 8]$.

Proof. Fix $\varepsilon > 0$ and let $\delta = \varepsilon/16$. Take $x, y \in [0, 8]$ with $|x - y| < \delta$. Then

$$|x^2 - y^2| = |x - y| |x + y| < \frac{\varepsilon}{16} |x + y| \leq \frac{\varepsilon}{16} \cdot 16 = \varepsilon,$$

since $0 \leq x + y \leq 16$ on $[0, 8]$. ■

Example 10.7 (Bounded is not enough). Is a continuous function on a *bounded* set automatically uniformly continuous? **No**. Consider $g(x) = \sin(1/x)$ on $A = (0, 1]$.



The oscillations near 0 become arbitrarily fast, so no single δ can control $|g(x) - g(y)|$ across the whole domain. A proof runs through the Sequential Criterion for Absence of Uniform Continuity (Abbott, Thm. 4.4.5).

Theorem 10.8 (Uniform Continuity on Compact Sets). *A function that is continuous on a compact set K is uniformly continuous on K .*

Remark 10.9. Abbott proves this with the sequential characterization of compactness (p. 133). An open-cover argument also works: for each $c \in K$ continuity supplies a δ_c ; the neighborhoods $V_{\delta_c/2}(c)$ cover K ; extract a finite subcover and take δ to be the smallest of the finitely many radii. Compactness is precisely what converts “a δ at each point” into “one δ for all points.”

Chapter 3: The Derivative

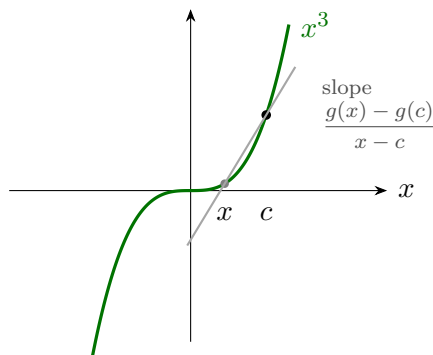
Derivatives are built on functional limits. Everything in this chapter descends from Definition 9.1, the Algebraic Limit Theorem, and — for the Mean Value Theorem — the Extreme Value Theorem of section 10.

11 The Derivative

Definition 11.1 (Derivative at a point). Let $g : A \rightarrow \mathbb{R}$ with A an interval, and let $c \in A$. The *derivative of g at c* is

$$g'(c) = \lim_{x \rightarrow c} \frac{g(x) - g(c)}{x - c},$$

provided the limit exists. If g is differentiable at every $c \in A$ we say g is *differentiable on A* and define the function $g' : A \rightarrow \mathbb{R}$, $c \mapsto g'(c)$.



Example 11.2 ($g(x) = x^3$). Compute $g'(c)$ from the definition. For $x \neq c$,

$$\frac{x^3 - c^3}{x - c} = \frac{(x - c)(x^2 + xc + c^2)}{x - c} = x^2 + xc + c^2,$$

where the cancellation is legitimate because Definition 9.1 only ever evaluates at $x \neq c$. Hence, by the Algebraic Limit Theorem,

$$g'(c) = \lim_{x \rightarrow c} (x^2 + xc + c^2) = \left(\lim_{x \rightarrow c} x \right)^2 + \left(\lim_{x \rightarrow c} x \right)c + c^2 = c^2 + c^2 + c^2 = 3c^2.$$

11.1 Continuity versus differentiability

Theorem 11.3 (Abbott 5.2.3). *If g is differentiable at c , then g is continuous at c . The converse is false.*

Example 11.4. $g(x) = |x|$ is continuous on $\mathbb{R}$ but not differentiable at 0: the difference quotient equals 1 for $x > 0$ and -1 for $x < 0$, so the limit does not exist.

Remark 11.5 (Is g' itself continuous?). Not necessarily. The standard example is

$$g(x) = \begin{cases} x^2 \sin(1/x) & x \neq 0, \\ 0 & x = 0. \end{cases}$$

This g is differentiable everywhere (including at 0, where $g'(0) = 0$ by a squeeze), but g' is not continuous at 0. See Abbott §5.1.

Notation 11.6 (Smoothness). “ g is C^k ” means $g^{(k)}$ exists and is continuous. The strongest condition, “ g is C^∞ ” (*smooth*), means all derivatives $g', g'', g''', \dots, g^{(k)}, \dots$ exist for every $k \in \mathbb{N}$.

11.2 Derivative rules

Theorem 11.7 (Algebraic Differentiability Theorem, Abbott 5.2.4). *Let f and g be defined on an interval A and differentiable at $c \in A$. Then*

- (i) $(f + g)'(c) = f'(c) + g'(c)$;
- (ii) $(kf)'(c) = kf'(c)$ for all $k \in \mathbb{R}$;

$$(iii) \quad (fg)'(c) = f'(c)g(c) + f(c)g'(c) \quad (\text{product rule});$$

$$(iv) \quad \left(\frac{f}{g}\right)'(c) = \frac{g(c)f'(c) - f(c)g'(c)}{[g(c)]^2}, \text{ provided } g(c) \neq 0 \quad (\text{quotient rule}).$$

Theorem 11.8 (Chain Rule, Abbott 5.2.5). *Let $f : A \rightarrow \mathbb{R}$ and $g : B \rightarrow \mathbb{R}$ satisfy $f(A) \subseteq B$, so that $g \circ f$ is defined. If f is differentiable at $c \in A$ and g is differentiable at $f(c) \in B$, then $g \circ f$ is differentiable at c with*

$$(g \circ f)'(c) = g'(f(c)) \cdot f'(c).$$

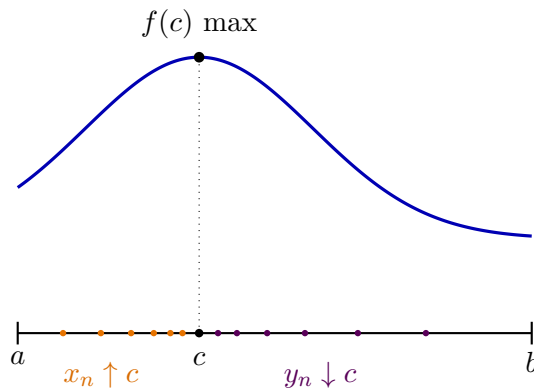
Scratch work. Write $d = f(c)$ and, for $x \neq c$, factor the difference quotient:

$$\frac{g(f(x)) - g(f(c))}{x - c} = \underbrace{\frac{g(f(x)) - g(f(c))}{f(x) - f(c)}}_{\rightarrow g'(d)} \cdot \underbrace{\frac{f(x) - f(c)}{x - c}}_{\rightarrow f'(c)}.$$

As $x \rightarrow c$ we have $y = f(x) \rightarrow f(c) = d$, since f is differentiable and hence continuous.

Wrinkle: the factorization divides by $f(x) - f(c)$, which may vanish at points $x \neq c$. Abbott's proof repairs this by replacing the first factor with an auxiliary function that is defined and continuous even where $f(x) = d$.

Theorem 11.9 (Interior Extremum Theorem). *If f is differentiable on the open interval (a, b) and f attains a maximum or minimum value at some $c \in (a, b)$, then $f'(c) = 0$.*



Proof. Suppose $f(c)$ is a maximum. Take sequences $(x_n) \rightarrow c$ with $x_n < c$ and $(y_n) \rightarrow c$ with $y_n > c$. Since $f(x_n) - f(c) \leq 0$ and $x_n - c < 0$,

$$\frac{f(x_n) - f(c)}{x_n - c} \geq 0 \quad \text{for all } n \in \mathbb{N},$$

while $f(y_n) - f(c) \leq 0$ and $y_n - c > 0$ give

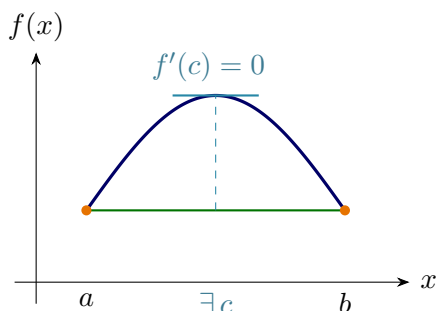
$$\frac{f(y_n) - f(c)}{y_n - c} \leq 0 \quad \text{for all } n \in \mathbb{N}.$$

By the Order Limit Theorem (Theorem 3.4) the two limits are ≥ 0 and ≤ 0 respectively. But f is differentiable at c , so by the sequential criterion for functional limits both limits equal $f'(c)$. Hence

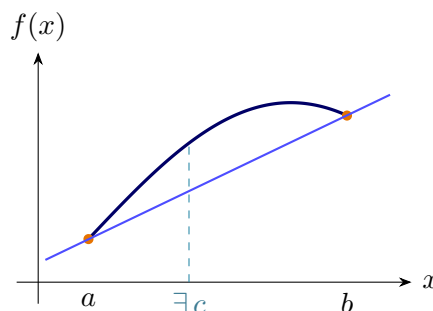
$$0 \leq f'(c) \leq 0,$$

so $f'(c) = 0$. The minimum case is symmetric. ■

12 The Mean Value Theorems



Rolle's Theorem, $f(a) = f(b)$



Mean Value Theorem

$$\text{slope} = \frac{f(b) - f(a)}{b - a}$$

Theorem 12.1 (Rolle's Theorem). *If $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then there exists a point $c \in (a, b)$ where $f'(c) = 0$.*

Proof. Because f is continuous on the compact set $[a, b]$, the Extreme Value Theorem (Theorem 10.3) says f achieves a maximum and a minimum value on $[a, b]$. Consider two cases.

Case 1: the max or the min occurs at some $c \in (a, b)$. Since f is differentiable on (a, b) , the Interior Extremum Theorem (Theorem 11.9) gives $f'(c) = 0$.

Case 2: both extreme values occur at the endpoints. Then $f(a) = f(b)$ is simultaneously the maximum and the minimum, so $f(x) = k$ is constant on $[a, b]$. Pick any $c \in (a, b)$; then

$$f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = \lim_{x \rightarrow c} \frac{k - k}{x - c} = \lim_{x \rightarrow c} 0 = 0.$$

In either case there is at least one $c \in (a, b)$ with $f'(c) = 0$. ■

Theorem 12.2 (Mean Value Theorem). *If $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b) , then there exists a point $c \in (a, b)$ where*

$$\underbrace{f'(c)}_{\text{instantaneous rate}} = \underbrace{\frac{f(b) - f(a)}{b - a}}_{\text{average (mean) rate}}.$$

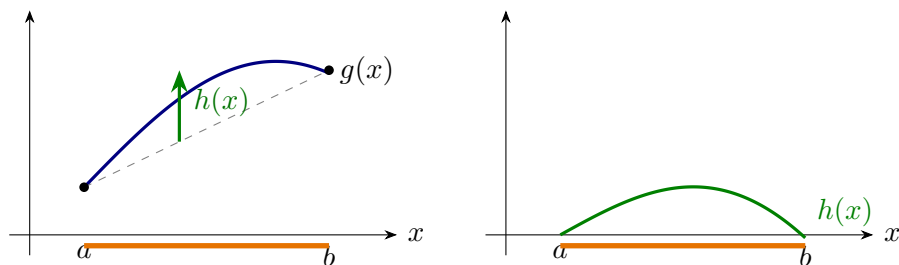
Proof. The strategy is to subtract off the secant line and apply Rolle's Theorem.

1. The secant line. Let g be the line through $(a, f(a))$ and $(b, f(b))$:

$$g(x) - f(a) = \frac{f(b) - f(a)}{b - a} (x - a) \quad \implies \quad g(x) = f(a) + \frac{f(b) - f(a)}{b - a} (x - a).$$

2. The gap function. Define $h : [a, b] \rightarrow \mathbb{R}$ by $h = f - g$, that is

$$h(x) = f(x) - \left(f(a) + \frac{f(b) - f(a)}{b - a} (x - a) \right).$$



3. Check Rolle's hypotheses. $h(a) = h(b) = 0$ by construction; h is continuous on $[a, b]$ because f is and g is linear (Theorem 9.10); and h is differentiable on (a, b) by Theorem 11.7.

4. Apply Rolle. There exists $c \in (a, b)$ with $h'(c) = 0$.

5. Compute h' .

$$h'(x) = \frac{d}{dx} \left[f(x) - f(a) - \frac{f(b) - f(a)}{b - a} (x - a) \right] = f'(x) - 0 - \frac{f(b) - f(a)}{b - a}.$$

6. Combine. At the point c from step 4,

$$0 = h'(c) = f'(c) - \frac{f(b) - f(a)}{b - a} \quad \implies \quad f'(c) = \frac{f(b) - f(a)}{b - a}. \quad \blacksquare$$

Remark 12.3 (Why these hypotheses?). Both are needed.

- *Differentiability on (a, b) :* for $f(x) = |x|$ on $[-1, 1]$ the average rate of change is 0, but f' is never 0.
- *Continuity on the closed $[a, b]$, not just (a, b) :* a function that jumps at an endpoint can have every interior slope miss the secant slope.

12.1 Corollaries: inferring f from f'

Corollary 12.4 (Abbott 5.3.3). *If $g : A \rightarrow \mathbb{R}$ is differentiable on an interval A and satisfies $g'(x) = 0$ for all $x \in A$, then $g(x) = k$ for some constant $k \in \mathbb{R}$.*

Proof. Choose $x_0 \in A$ and let $k = g(x_0)$. Let $y \in A$ be another point; without loss of generality $y > x_0$. Since g is differentiable, it is continuous on $[x_0, y] \subseteq A$, so the Mean Value Theorem supplies $c \in (x_0, y)$ with

$$g'(c) = \frac{g(y) - g(x_0)}{y - x_0}.$$

But $g'(c) = 0$, so $g(y) = g(x_0) = k$. Since y was arbitrary, $g \equiv k$ on A . ■

Corollary 12.5. *Let f, g be differentiable on an interval A .*

- *If $f'(x) = g'(x)$ for all $x \in A$, then $f(x) = g(x) + k$ for some constant k . (Apply Corollary 12.4 to $h = f - g$.)*
- *If $f'(x) = f(x)$ for all $x \in A$, then $f(x) = ke^x$. (Differentiate $f(x)e^{-x}$ and apply Corollary 12.4.)*
- *If $g'(c) > 0$ for all $c \in A$, then g is strictly increasing on A : $x < y$ implies $g(x) < g(y)$.*

Chapter 4: Sequences and Series of Functions

We now iterate the constructions of Chapter 1 one level up: instead of a sequence of *numbers* (a_n) , we consider a sequence of *functions* (f_n) . The central question of the chapter is inheritance — if $f_n \rightarrow f$, which properties of the f_n survive in f ? The answer turns on the distinction between pointwise and uniform convergence.

13 Sequences of Functions

Sequence of real numbers	Sequence of functions
$(a_1, a_2, a_3, \dots) = (a_n)_{n=1}^\infty$	$(f_1, f_2, f_3, \dots) = (f_n)_{n=1}^\infty$
$a_n \in \mathbb{R}$	$f_n : A \rightarrow \mathbb{R}$

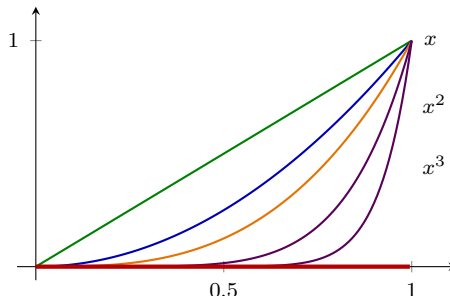
Remark 13.1 (Why study these?). Two reasons. First, they prepare us for *series* of functions. Second, they give “infinite polynomial” representations of familiar functions — e.g.

$$e^x = 1 + x + \frac{1}{2}x^2 + \frac{1}{3!}x^3 + \dots$$

— and polynomials are the functions we understand best.

13.1 Pointwise convergence

Definition 13.2 (Pointwise convergence). A sequence of functions (f_n) , $f_n : A \rightarrow \mathbb{R}$, converges *pointwise* on A to $f : A \rightarrow \mathbb{R}$ if for every $x \in A$ the sequence of real numbers $(f_n(x))_{n=1}^{\infty}$ converges to $f(x)$. We write $f_n \xrightarrow{\text{pw}} f$.



Example 13.3 ($f_n(x) = x^n$ on $[0, 1]$).

For $x \in [0, 1)$ we have $\lim_{n \rightarrow \infty} x^n = 0$, while for $x = 1$, $\lim_{n \rightarrow \infty} 1^n = 1$. Hence $f_n \xrightarrow{\text{pw}} f$ where

$$f(x) = \begin{cases} 0 & x \in [0, 1), \\ 1 & x = 1. \end{cases}$$

Each f_n is continuous; the limit f is *not*.

Example 13.4. Find the pointwise limit of $f_n(x) = \frac{nx^2}{1 + nx^2}$ on $\mathbb{R}$.

For $x \neq 0$ divide through by nx^2 :

$$f_n(x) = \frac{1}{\frac{1}{nx^2} + 1}, \quad \text{so} \quad \lim_{n \rightarrow \infty} f_n(x) = \frac{1}{\left(\lim_{n \rightarrow \infty} \frac{1}{n}\right) \frac{1}{x^2} + 1} = \frac{1}{0 + 1} = 1$$

by the Algebraic Limit Theorem. For $x = 0$, $f_n(0) = 0/1 = 0$ for every n , so the limit is 0. Therefore

$$f(x) = \begin{cases} 1 & x \neq 0, \\ 0 & x = 0. \end{cases}$$

Remark 13.5. Both examples feature sequences of *continuous* functions whose pointwise limit is *discontinuous*. This motivates the question: under what conditions on the convergence does f inherit the properties of the f_n ?

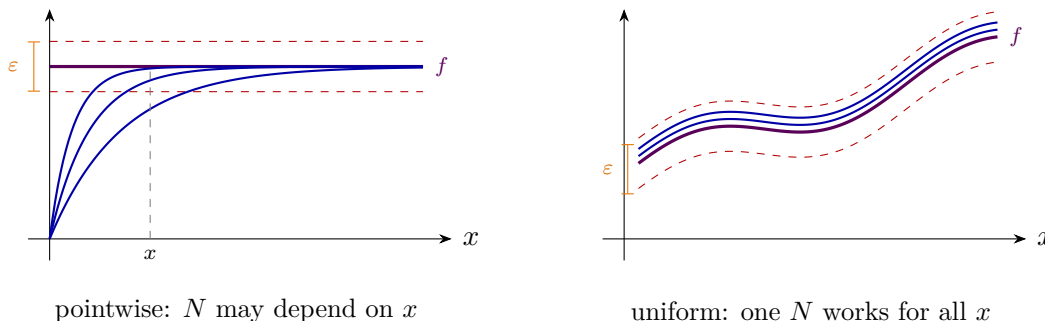
13.2 Uniform convergence

Definition 13.6 (Uniform convergence). A sequence (f_n) converges *uniformly* to f on A if

$$\forall \varepsilon > 0 \quad \exists N \in \mathbb{N} \text{ such that } \forall x \in A : \quad n \geq N \implies |f_n(x) - f(x)| < \varepsilon.$$

We write $f_n \xrightarrow{\text{unif}} f$.

The distinction from Definition 13.2 is once again the placement of a quantifier. In pointwise convergence, one first fixes x and then chooses N_x , which may depend on x ; in uniform convergence a single N must work for every $x \in A$ simultaneously. Geometrically: from index N onward, the entire graph of f_n lies inside an ε -tube around the graph of f .



Example 13.7 (x^n does not converge uniformly on $[0, 1]$). Two arguments.

Via the Continuous Limit Theorem. Each x^n is continuous on $[0, 1]$; if the convergence were uniform, Theorem 13.9 below would force the limit f to be continuous. It is not, so the convergence is not uniform.

Directly. Negating Definition 13.6: there exists $\varepsilon > 0$ such that for every $N \in \mathbb{N}$ there is some $x \in A$ and some $n \geq N$ with $|f_n(x) - f(x)| \geq \varepsilon$. Take $\varepsilon = \frac{1}{2}$. Given $N \in \mathbb{N}$, choose

$$x = \sqrt[N]{\frac{1}{2}} \in (0, 1).$$

Then with $n = N$,

$$|x^N - 0| = \frac{1}{2} \geq \frac{1}{2} = \varepsilon.$$

Example 13.8 (But it *does* converge uniformly on $[0, \frac{1}{2}]$). On $A = [0, 0.5]$ the limit is $f \equiv 0$. For every $n \in \mathbb{N}$ and every $x \in [0, 0.5]$, monotonicity of $t \mapsto t^n$ gives

$$|f_n(x) - f(x)| = x^n \leq (0.5)^n = |f_n(0.5) - f(0.5)|. \quad (*)$$

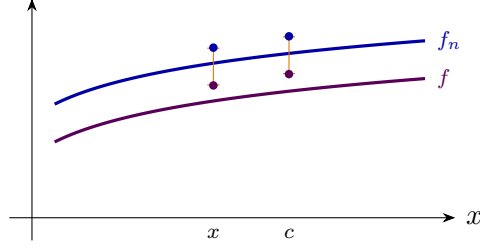
Fix $\varepsilon > 0$. Since $\lim_{n \rightarrow \infty} (0.5)^n = 0$, there is $N \in \mathbb{N}$ with $|(0.5)^n - 0| < \varepsilon$ whenever $n \geq N$. By (*) this same N works for every $x \in [0, 0.5]$, which is exactly Definition 13.6.

Theorem 13.9 (Continuous Limit Theorem). *If $f_n \xrightarrow{\text{unif}} f$ on A and each f_n is continuous at $c \in A$, then f is continuous at c .*

Scratch work. We want $|f(x) - f(c)|$ small. Insert f_n twice and use the triangle inequality:

$$|f(x) - f(c)| \leq \underbrace{|f(x) - f_n(x)|}_{\text{uniform convergence}} + \underbrace{|f_n(x) - f_n(c)|}_{\text{continuity of } f_n} + \underbrace{|f_n(c) - f(c)|}_{\text{uniform convergence}}.$$

Make each piece $< \varepsilon/3$. Note that the first and third terms need *uniform* convergence: the same index N must work at x and at c .



Proof. Fix $\varepsilon > 0$. Since $f_n \rightarrow f$ uniformly on A , there exists $N \in \mathbb{N}$ such that

$$|f_N(a) - f(a)| < \frac{\varepsilon}{3} \quad \text{for all } a \in A.$$

Because f_N is continuous at c , there exists $\delta > 0$ such that

$$|x - c| < \delta \implies |f_N(x) - f_N(c)| < \frac{\varepsilon}{3}.$$

Then for every $x \in A$ with $|x - c| < \delta$, the triangle inequality gives

$$|f(x) - f(c)| \leq |f(x) - f_N(x)| + |f_N(x) - f_N(c)| + |f_N(c) - f(c)| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \quad \blacksquare$$

13.3 The Cauchy criterion and the Differentiable Limit Theorem

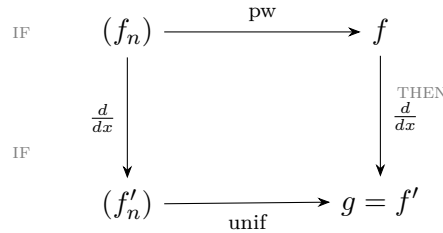
Theorem 13.10 (Cauchy Criterion for Uniform Convergence, Abbott 6.2.5). *A sequence (f_n) converges uniformly on A if and only if*

$$\forall \varepsilon > 0 \quad \exists N \in \mathbb{N} \text{ such that } \forall x \in A: \quad m, n \geq N \implies |f_m(x) - f_n(x)| < \varepsilon.$$

Theorem 13.11 (Differentiable Limit Theorem). *Suppose*

- $f_n \rightarrow f$ pointwise on $[a, b]$,
- each f_n is differentiable on $[a, b]$, and
- (f'_n) converges uniformly on $[a, b]$ to some function g .

Then f is differentiable on $[a, b]$ and $f' = g$.



Proof of the key estimate. The delicate step controls $\frac{f_m(x) - f_m(c)}{x - c} - \frac{f_n(x) - f_n(c)}{x - c}$. Apply the Mean Value Theorem (Theorem 12.2) to $f_m - f_n$ on $[c, x]$: this is legitimate because f_m and f_n are differentiable, hence continuous, on $[a, b] \supseteq [c, x]$. So there exists $\alpha \in (c, x)$ with

$$[f_m - f_n]'(\alpha) = \frac{(f_m(x) - f_n(x)) - (f_m(c) - f_n(c))}{x - c}.$$

By the Cauchy criterion (Theorem 13.10) applied to the uniformly convergent sequence (f'_n) , there is $N_2 \in \mathbb{N}$ such that $m, n \geq N_2$ implies $|f'_m(\hat{x}) - f'_n(\hat{x})| < \varepsilon/3$ for *all* $\hat{x} \in [a, b]$ — in particular at $\hat{x} = \alpha$. Hence for $m, n \geq N_2$,

$$\left| \frac{f_m(x) - f_m(c)}{x - c} - \frac{f_n(x) - f_n(c)}{x - c} \right| = |f'_m(\alpha) - f'_n(\alpha)| < \frac{\varepsilon}{3}.$$

Letting $m \rightarrow \infty$ and using $f_m \rightarrow f$ pointwise,

$$\left| \frac{f(x) - f(c)}{x - c} - \frac{f_n(x) - f_n(c)}{x - c} \right| \leq \frac{\varepsilon}{3}.$$

A final $\varepsilon/3$ argument, combining this with the differentiability of f_n at c and with $f'_n(c) \rightarrow g(c)$, gives $\left| \frac{f(x) - f(c)}{x - c} - g(c) \right| < \varepsilon$, i.e. $f'(c) = g(c)$. ■

14 Series of Functions

Definition 14.1 (Convergence of a series of functions). Given $f_n : A \rightarrow \mathbb{R}$ for each $n \in \mathbb{N}$, the infinite series

$$\sum_{n=1}^{\infty} f_n(x) = f_1(x) + f_2(x) + f_3(x) + \cdots$$

converges pointwise (respectively *uniformly*) to f on A if the sequence of partial sums

$$s_k(x) = f_1(x) + f_2(x) + \cdots + f_k(x)$$

converges pointwise (respectively uniformly) to f on A .

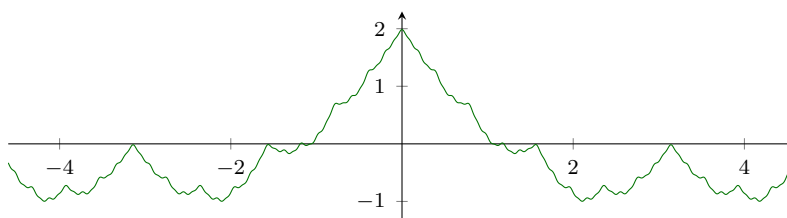
Example 14.2 (The geometric series). $f(x) = \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$ for $x \in (-1, 1)$; the domain of f is exactly the set where the series converges. Here we are lucky: the limit function is recognizable, so continuity and differentiability on $(-1, 1)$ can be read off directly — for instance

$$f'(x) = \frac{d}{dx} [(1-x)^{-1}] = (-1)(1-x)^{-2}(-1) = \frac{1}{(1-x)^2}.$$

Remark 14.3 (The central problem). Usually we are not so lucky. The task is to understand properties — continuity, differentiability — of $f(x) = \sum_{n=1}^{\infty} f_n(x)$ *even when we have no closed formula for f* .

Example 14.4 (Weierstrass's function). Let

$$g(x) = \sum_{n=0}^{\infty} \frac{\cos(2^n x)}{2^n}, \quad x \in \mathbb{R}.$$



Where does it converge? For all $x \in \mathbb{R}$: since $\left| \frac{\cos(2^n x)}{2^n} \right| \leq \frac{1}{2^n}$ and $\sum 1/2^n$ converges, the comparison test gives absolute convergence, hence convergence, at every real x . Remarkably, g is continuous on all of $\mathbb{R}$ but differentiable *nowhere* (Abbott §5.4).

Theorem 14.5 (Cauchy Criterion for Uniform Convergence of a Series). $\sum_{n=1}^{\infty} f_n$ converges uniformly on A if and only if

$$\forall \varepsilon > 0 \quad \exists N \in \mathbb{N} \text{ such that } \forall x \in A : \quad m, n \geq N \implies |f_{m+1}(x) + \cdots + f_n(x)| < \varepsilon.$$

Where this comes from. Apply Theorem 13.10 to the sequence of partial sums $s_n = \sum_{k=1}^n f_k$ and note that, for $n > m$,

$$s_n(x) - s_m(x) = (f_1(x) + \cdots + f_n(x)) - (f_1(x) + \cdots + f_m(x)) = f_{m+1}(x) + \cdots + f_n(x). \quad \blacksquare$$

Corollary 14.6 (Weierstrass M -Test). Suppose that for each $n \in \mathbb{N}$ we have $f_n : A \rightarrow \mathbb{R}$ and a real number $M_n > 0$ satisfying

$$|f_n(x)| \leq M_n \quad \text{for all } x \in A.$$

If $\sum_{n=1}^{\infty} M_n$ converges, then $\sum_{n=1}^{\infty} f_n$ converges uniformly on A .

Example 14.7 (Weierstrass's function is continuous). The proof is a three-step template worth memorizing.

1. **The partial sums are continuous.** For each N ,

$$s_N(x) = \sum_{n=0}^N \frac{\cos(2^n x)}{2^n} = \frac{\cos(2^0 x)}{2^0} + \frac{\cos(2^1 x)}{2^1} + \cdots + \frac{\cos(2^N x)}{2^N}$$

is a *finite* sum of continuous functions, hence continuous by the Algebraic Continuity Theorem (Theorem 9.10).

2. **The series converges uniformly.** With $f_n(x) = \cos(2^n x)/2^n$ we have $|f_n(x)| \leq 1/2^n =: M_n$ for all $x \in \mathbb{R}$, and $\sum 1/2^n$ converges (geometric, ratio $\frac{1}{2} < 1$). By the Weierstrass M -Test (Corollary 14.6), the series converges uniformly on $\mathbb{R}$.

3. **Apply the Continuous Limit Theorem.** The partial sums are continuous and converge uniformly on $\mathbb{R}$, so by Theorem 13.9 the limit function g is continuous on $\mathbb{R}$.

Example 14.8 (The Riemann zeta function). Another function defined by a series:

$$\zeta(x) = \sum_{n=1}^{\infty} \frac{1}{n^x}.$$

Its natural domain is $x > 1$, i.e. $(1, \infty)$; the same three-step template shows it is continuous there. (Extending ζ beyond this domain — via analytic continuation, giving the notorious $\zeta(-1) = -\frac{1}{12}$ — belongs to complex analysis.)

15 Power Series

Definition 15.1 (Power series). A *power series* is a series of the form

$$\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots, \quad a_n \in \mathbb{R}.$$

(At $x = 0$ the sum is a_0 .)

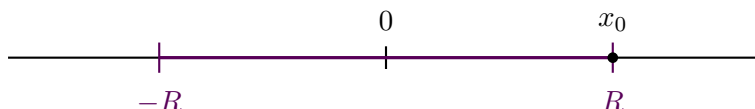
Example 15.2 (Which of these are power series?).

- $\sum_{k=0}^{\infty} \frac{x^k}{k!}$ — **yes**, with $a_k = 1/k!$.

- $1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$ — **yes**; $a_n = 0$ for odd n .
- $\sum_{n=0}^{\infty} \sin(x) x^n$ — **no**: the coefficient $\sin(x)$ depends on x .
- $\sum_{n=0}^{1000} n^2 x^n$ — not an infinite sum as written, though one may set $a_n = 0$ for $n > 1000$ to view it as one.

Remark 15.3 (Why power series are special). They are a restricted class of series — entirely determined by the coefficients (a_n) — and consequently enjoy much stronger theorems than general series of functions.

Theorem 15.4 (Basic Fact 1). If $\sum_{n=0}^{\infty} a_n x^n$ converges at some $x_0 \neq 0$, then it converges pointwise on $(-R, R)$, where $R = |x_0|$.



Remark 15.5. This “all-or-nothing on a symmetric interval” behaviour is special to power series. The series $\sum_{n=1}^{\infty} n^{-x}$ of Example 14.8, for example, converges precisely on $(1, \infty)$ — not on an interval centred at 0.

Theorem 15.6 (Basic Fact 2). Either $\sum a_n x^n$ converges only at $x = 0$, or there exists $R > 0$ (the radius of convergence) such that $\sum a_n x^n$ converges uniformly on every compact interval $[-r, r]$ with $0 < r < R$.



Proof. Assume the series converges at some $x_0 \neq 0$ and fix $0 < r < R$ where $R = |x_0|$; we show uniform convergence on $[-r, r]$ via the M -test.

Since $\sum a_n x_0^n$ converges, its terms tend to zero, so $(a_n x_0^n)$ is bounded: there is $Q > 0$ with $|a_n| |x_0|^n = |a_n| R^n \leq Q$ for all n . Now for $x \in [-r, r]$, so that $|x| \leq r$,

$$|a_n x^n| = |a_n| |x|^n \leq |a_n| r^n = |a_n| R^n \left(\frac{r}{R}\right)^n \leq Q \rho^n =: M_n, \quad \rho = \frac{r}{R} \in (0, 1).$$

Then $\sum M_n = Q \sum \rho^n$ is a convergent geometric series, so by the Weierstrass M -Test the power series converges uniformly on $[-r, r]$. ■

Theorem 15.7 (Basic Fact 3). *Suppose $\sum_{n=0}^{\infty} a_n x^n$ converges pointwise to $f(x)$ on an interval $(-R, R)$. Then:*

- f is continuous on $(-R, R)$;
- f is differentiable any number of times on $(-R, R)$ — that is, $f \in C^\infty$; and
- the derivatives may be computed term by term:

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}, \quad f''(x) = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}, \quad \dots$$

with each of these series converging uniformly on $[-r, r]$ for $0 < r < R$.

Proof idea for term-by-term differentiation. Step 1. Show $\sum_{n \geq 1} n a_n x^{n-1}$ converges uniformly on $[-r, r]$ by the M -test. Reusing the bound from Theorem 15.6,

$$|n a_n x^{n-1}| = n |a_n| |x|^{n-1} \leq n |a_n| r^{n-1} = \frac{n |a_n| r^n}{r} \leq \frac{Q}{r} n \rho^n,$$

and $\sum n \rho^n$ converges because exponential decay beats linear growth.

Step 2. Apply the Differentiable Limit Theorem (Theorem 13.11) to the sequence of partial sums: the polynomials $s_N(x) = a_0 + a_1 x + \dots + a_N x^N$ converge pointwise on $(-R, R)$ to f , and their derivatives $s'_N(x) = a_1 + 2a_2 x + \dots + N a_N x^{N-1}$ converge uniformly on $[-r, r]$. Hence f is differentiable with $f' = \lim s'_N = \sum n a_n x^{n-1}$. Induction gives all higher derivatives. ■

Example 15.8 (Recognizing a power series). Does $f(x) = \sum_{n=1}^{\infty} \frac{1}{n} x^n = x + \frac{1}{2} x^2 + \frac{1}{3} x^3 + \dots$ converge to a function we know? Differentiating term by term (Theorem 15.7),

$$f'(x) = 1 + x + x^2 + \dots = \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}, \quad x \in (-1, 1),$$

so $f(x) = -\log(1-x) + C$; and $f(0) = 0$ forces $C = 0$. Hence

$$\log(1-x) = -x - \frac{1}{2} x^2 - \frac{1}{3} x^3 - \dots, \quad \log(1+x) = x - \frac{1}{2} x^2 + \frac{1}{3} x^3 - \dots.$$

16 Taylor Series

Remark 16.1 (One equation, two perspectives). The identity $f(x) = \sum_{n=0}^{\infty} a_n x^n$ can be read from either side.

- *Power series*: start with the infinite sum. Does it converge, and where? Is the function it defines continuous, differentiable, recognizable?
- *Taylor series*: start with a function f , say $\sin x$. Can it be expressed as a power series? If so, what must the coefficients be? And for which x does the series actually converge back to f ?

16.1 Deriving the coefficients

Suppose $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \dots$ on some interval. By Theorem 15.7 we may differentiate term by term repeatedly and evaluate at 0:

$$\begin{aligned} f(x) &= a_0 + a_1x + a_2x^2 + a_3x^3 + \dots &\implies& f(0) = a_0, \\ f'(x) &= a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + \dots &\implies& f'(0) = a_1, \\ f''(x) &= 2a_2 + 3 \cdot 2 a_3x + 4 \cdot 3 a_4x^2 + \dots &\implies& f''(0) = 2a_2, \\ f^{(3)}(x) &= 3 \cdot 2 a_3 + 4 \cdot 3 \cdot 2 a_4x + \dots &\implies& f^{(3)}(0) = 3 \cdot 2 a_3. \end{aligned}$$

Solving gives $a_0 = f(0)$, $a_1 = f'(0)/1!$, $a_2 = f''(0)/2!$, $a_3 = f^{(3)}(0)/3!$, and in general

$$a_n = \frac{f^{(n)}(0)}{n!}$$

Definition 16.2 (Taylor series). If f is infinitely differentiable on an interval about 0, its *Taylor series* (about 0) is

$$f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots$$

The derivation above shows only this: *if* a power series representation exists, these must be its coefficients.

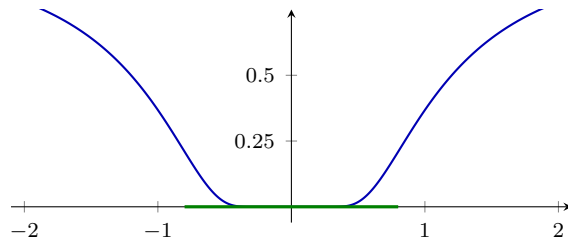
16.2 The good, the bad, and the ugly

Given a C^∞ function f , its Taylor series may:

- **converge to f** on some interval — the good case;
- **converge, but to a different function** — the bad case;
- **diverge everywhere except $x = 0$** — the ugly case.

Example 16.3 (The bad case). Let

$$f(x) = \begin{cases} e^{-1/x^2} & x \neq 0, \\ 0 & x = 0. \end{cases}$$



One can check that f is continuous at 0, differentiable any number of times at 0, and that $f^{(n)}(0) = 0$ for every n . Its Taylor series is therefore

$$f(0) + f'(0)x + \frac{f''(0)}{2}x^2 + \cdots = 0 + 0 + 0 + \cdots = 0.$$

The series converges everywhere — but to the zero function, which is *not* f .

Example 16.4 (The ugly case). A theorem of Émile Borel says that for *any* sequence (b_n) there is a smooth f with $f^{(n)}(0) = b_n$. Take $b_n = (n!)^2$; then

$$a_n = \frac{f^{(n)}(0)}{n!} = \frac{(n!)^2}{n!} = n!,$$

and the Taylor series $0! + 1!x + 2!x^2 + 3!x^3 + \cdots$ diverges for every $x \neq 0$.

16.3 Remainder theorems

To certify the good case we must show the truncation error vanishes.

Definition 16.5 (Taylor polynomial and error term). The n th Taylor polynomial of f is

$$S_n(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n,$$

and the n th error term is $E_n(x) = f(x) - S_n(x)$, so that $f(x) = S_n(x) + E_n(x)$. If $E_n(x) \rightarrow 0$ as $n \rightarrow \infty$, then $S_n(x) \rightarrow f(x)$, i.e. the Taylor series converges to f at x .

Theorem 16.6 (Lagrange Remainder Theorem). *If f is $(n+1)$ -times differentiable on $(-R, R)$, then for each $x \in (-R, R)$ there exists $c \in \mathbb{R}$ with $|c| < |x|$ such that*

$$E_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} x^{n+1}.$$

Remark 16.7. The proof uses the Generalized Mean Value Theorem. The statement is a direct descendant of Theorem 12.2: the case $n = 0$ reads $f(x) - f(0) = f'(c)x$.

Example 16.8 ($\sin x$ equals its Taylor series on all of $\mathbb{R}$). Fix $x \in \mathbb{R}$. By Theorem 16.6 there is c with $|c| < |x|$ and

$$|E_n(x)| = \left| \frac{f^{(n+1)}(c)}{(n+1)!} x^{n+1} \right| = \frac{|f^{(n+1)}(c)|}{(n+1)!} |x|^{n+1} \leq \frac{|x|^{n+1}}{(n+1)!},$$

since every derivative of $\sin$ is $\pm \sin$ or $\pm \cos$, all bounded by 1. By Lemma 16.9 below the right-hand side tends to 0, so $0 \leq |E_n(x)| \rightarrow 0$ and the Squeeze Theorem gives $E_n(x) \rightarrow 0$.

Lemma 16.9 (Factorial growth beats exponential). *For every $x \in \mathbb{R}$, the sequence $\left(\frac{x^n}{n!}\right)_{n=1}^{\infty}$ converges to 0.*

Proof. Fix $\varepsilon > 0$. Choose $N_1 \in \mathbb{N}$ with $N_1 > |x|$, so that $|x|/N_1 < 1$. The sequence $\left(|x|/N_1\right)^n$ is geometric with ratio in $(0, 1)$, hence converges to 0; so there exists $N_2 \in \mathbb{N}$ with

$$\left(\frac{|x|}{N_1}\right)^{N_2} < \frac{\varepsilon}{|x|^{N_1}/N_1!}.$$

Let $N = N_1 + N_2$. Then for $n \geq N$,

$$\left|\frac{x^n}{n!} - 0\right| = \frac{|x|}{1} \cdot \frac{|x|}{2} \cdots \frac{|x|}{n} = \underbrace{\frac{|x|^{N_1}}{N_1!}}_{\text{fixed}} \cdot \underbrace{\frac{|x|}{N_1+1} \cdots \frac{|x|}{n}}_{\text{each factor} < |x|/N_1} \leq \frac{|x|^{N_1}}{N_1!} \left(\frac{|x|}{N_1}\right)^{n-N_1} \leq \frac{|x|^{N_1}}{N_1!} \left(\frac{|x|}{N_1}\right)^{N_2} < \varepsilon.$$

■

Chapter 5: The Riemann Integral

Four questions organize this chapter.

- Which functions can we integrate?
- What properties does the integral have?
- If $f_n \rightarrow f$, does $\int f_n \rightarrow \int f$?
- When does the Fundamental Theorem of Calculus, $\int_a^b F' = F(b) - F(a)$, hold?

Example 16.10 (Predict: which of these are integrable on $[-1, 1]$?).

Function	Integrable?
Heaviside $H(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases}$	yes (Example 18.1)
Weierstrass $h(x) = \sum_{n=0}^{\infty} \frac{\cos(2^n x)}{2^n}$	yes (continuous)
Dirichlet $f(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ 0 & x \notin \mathbb{Q} \end{cases}$	no (Example 18.4)
Thomae $t(x)$ of Example 9.9	yes (Example 18.5)

17 Definitions: Partitions, Sums, and Integrals

Throughout, f is a *bounded* function on $[a, b]$.

Definition 17.1 (Partition, upper and lower sums). A *partition* of $[a, b]$ is a finite subset of $[a, b]$ containing both a and b , written

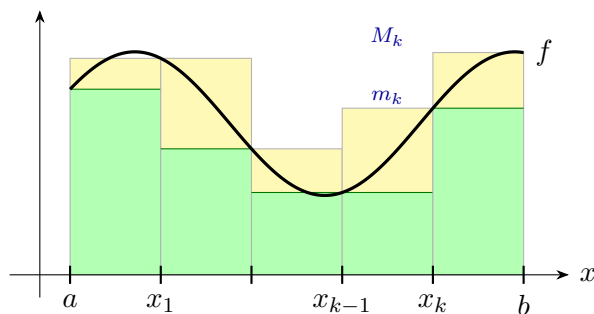
$$P = \{a = x_0, x_1, x_2, \dots, x_n = b\}.$$

For each $k = 1, \dots, n$ set

$$m_k = \inf\{f(x) : x \in [x_{k-1}, x_k]\}, \quad M_k = \sup\{f(x) : x \in [x_{k-1}, x_k]\}.$$

The *lower sum* and *upper sum* of f over P are

$$L(f, P) = \sum_{k=1}^n m_k (x_k - x_{k-1}), \quad U(f, P) = \sum_{k=1}^n M_k (x_k - x_{k-1}).$$



Definition 17.2 (Refinement). A partition Q is a *refinement* of a partition P if $P \subseteq Q$ (that is, Q contains every point of P , and possibly more).

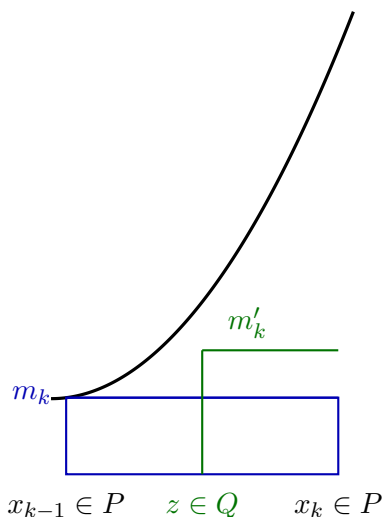
Lemma 17.3. If $P \subseteq Q$ then

$$L(f, P) \leq L(f, Q) \quad \text{and} \quad U(f, P) \geq U(f, Q).$$

Proof sketch. It suffices to insert one new point $z \in (x_{k-1}, x_k)$. Comparing the contributions to the lower sums, and writing m'_k for the infimum over $[z, x_k]$,

$$m_k(z - x_{k-1}) + m'_k(x_k - z) \geq m_k(z - x_{k-1}) + m_k(x_k - z) = m_k(x_k - x_{k-1}),$$

because taking the infimum over a smaller interval can only increase it. So the lower sum does not decrease. Upper sums are symmetric. ■



Lemma 17.4. For any two partitions P_1, P_2 of $[a, b]$ we have $L(f, P_1) \leq U(f, P_2)$.

Proof. Let $Q = P_1 \cup P_2$, the common refinement of P_1 and P_2 . Then

$$L(f, P_1) \underset{\text{Lemma 17.3}}{\leq} L(f, Q) \underset{\text{definitions}}{\leq} U(f, Q) \underset{\text{Lemma 17.3}}{\leq} U(f, P_2). \quad \blacksquare$$

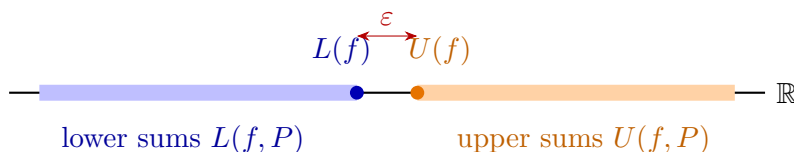
Definition 17.5 (Upper and lower integrals). Let $\mathbb{P}$ denote the (uncountably infinite) set of all partitions of $[a, b]$. The *upper* and *lower integrals* of f are

$$U(f) = \inf\{U(f, P) : P \in \mathbb{P}\}, \quad L(f) = \sup\{L(f, P) : P \in \mathbb{P}\}.$$

By Lemma 17.4, $L(f) \leq U(f)$.

Definition 17.6 (Riemann integrability). A bounded function f on $[a, b]$ is *integrable* if $L(f) = U(f)$, in which case we write

$$\int_a^b f = L(f) = U(f).$$



Theorem 17.7 (ε -Criterion for Integrability, Abbott 7.2.8). A bounded function f is integrable on $[a, b]$ if and only if

$$\forall \varepsilon > 0 \quad \exists \text{ a partition } P_\varepsilon \text{ of } [a, b] \text{ such that } U(f, P_\varepsilon) - L(f, P_\varepsilon) < \varepsilon.$$

This criterion — not Definition 17.6 — is the practical tool. Nearly every proof below is an exercise in exhibiting a good partition.

18 Which Functions Are Integrable?

Example 18.1 (The Heaviside function). Show that $H(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases}$ is integrable on $[-1, 2]$.

Scratch work. The obvious partition $P = \{-1, 0, 2\}$ gives $L(H, P) = 0 \cdot 1 + 1 \cdot 2 = 2$ and $U(H, P) = 0 \cdot 1 + 1 \cdot 2 = 2$ — wait: on $[-1, 0]$ the supremum of H is $H(0) = 1$, not 0. So $U(H, P) = 1 \cdot 1 + 1 \cdot 2 = 3 \neq 2$. The fix is to isolate the jump inside a thin sliver.

Proof. Fix $\varepsilon > 0$ and take the partition

$$P_\varepsilon = \left\{-1, -\frac{\varepsilon}{3}, \frac{\varepsilon}{3}, 2\right\}.$$

On $[-1, -\varepsilon/3]$ both inf and sup of H equal 0; on $[-\varepsilon/3, \varepsilon/3]$ the infimum is 0 and the supremum is 1; on $[\varepsilon/3, 2]$ both equal 1. Hence

$$L(H, P_\varepsilon) = 0 + 0 + 1 \cdot \left(2 - \frac{\varepsilon}{3}\right), \quad U(H, P_\varepsilon) = 0 + 1 \cdot \frac{2\varepsilon}{3} + 1 \cdot \left(2 - \frac{\varepsilon}{3}\right),$$

so

$$U(H, P_\varepsilon) - L(H, P_\varepsilon) = \frac{2\varepsilon}{3} < \varepsilon.$$

By Theorem 17.7, H is integrable on $[-1, 2]$ (with $\int_{-1}^2 H = 2$). ■

Theorem 18.2 (Continuous $\Rightarrow$ integrable). *If f is continuous on $[a, b]$, then f is integrable on $[a, b]$.*

Scratch work. We need a partition making every $M_k - m_k$ small. Uniform continuity supplies exactly that: it gives one δ controlling the oscillation of f on every subinterval of width $< \delta$ at once.

Proof. Since $[a, b]$ is compact and f is continuous, the Extreme Value Theorem (Theorem 10.3) shows f is bounded on $[a, b]$, so upper and lower sums are defined.

Fix $\varepsilon > 0$. By Theorem 10.8, f is in fact *uniformly* continuous on the compact set $[a, b]$, so there exists $\delta > 0$ with

$$|x - y| < \delta \implies |f(x) - f(y)| < \frac{\varepsilon}{b - a}.$$

Let $P = \{a = x_0, x_1, \dots, x_n = b\}$ be any partition with $x_k - x_{k-1} < \delta$ for each k . Then

$$U(f, P) - L(f, P) = \sum_{k=1}^n M_k(x_k - x_{k-1}) - \sum_{k=1}^n m_k(x_k - x_{k-1}) = \sum_{k=1}^n (M_k - m_k)(x_k - x_{k-1}).$$

By the Extreme Value Theorem again, on each compact subinterval $[x_{k-1}, x_k]$ the continuous function f attains its supremum and infimum: $M_k = f(y_k)$ and $m_k = f(z_k)$ for some y_k, z_k in that subinterval. Since $|y_k - z_k| < \delta$ by construction,

$$M_k - m_k = |f(y_k) - f(z_k)| < \frac{\varepsilon}{b - a}.$$

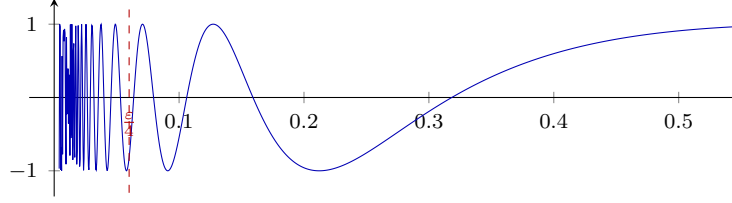
Therefore

$$U(f, P) - L(f, P) < \frac{\varepsilon}{b - a} \sum_{k=1}^n (x_k - x_{k-1}) = \frac{\varepsilon}{b - a} (b - a) = \varepsilon,$$

the sum telescoping to $x_n - x_0 = b - a$. Apply Theorem 17.7. ■

Example 18.3 (The topologist's sine curve is integrable on $[0, 1]$). Let

$$f(x) = \begin{cases} \sin(1/x) & x \neq 0, \\ 0 & x = 0. \end{cases}$$



Proof. Fix $\varepsilon > 0$. On $[\varepsilon/4, 1]$ the function f is continuous, hence integrable by Theorem 18.2; by Theorem 17.7 there is a partition $P = \{\varepsilon/4 = x_1, x_2, \dots, x_n = 1\}$ of $[\varepsilon/4, 1]$ with

$$U(f, P) - L(f, P) < \frac{\varepsilon}{2}.$$

Let $Q = \{0\} \cup P$, a partition of $[0, 1]$. Then

$$U(f, Q) - L(f, Q) = \sum_{k=1}^n (M_k - m_k)(x_k - x_{k-1}) = \underbrace{(M_1 - m_1) \left(\frac{\varepsilon}{4} - 0\right)}_{(I)} + \underbrace{\sum_{k=2}^n (M_k - m_k)(x_k - x_{k-1})}_{(II)}.$$

Term (I) equals $(1 - (-1))\frac{\varepsilon}{4} = \frac{\varepsilon}{2}$, since $\sin(1/x)$ takes values arbitrarily close to 1 and -1 in every neighborhood of 0. Term (II) is exactly $U(f, P) - L(f, P) < \varepsilon/2$. Hence $U(f, Q) - L(f, Q) < \varepsilon$. ■

Example 18.4 (Dirichlet's function is *not* integrable on $[0, 1]$). Let $f(x) = 1$ for $x \in \mathbb{Q}$ and $f(x) = 0$ for $x \notin \mathbb{Q}$. Take $\varepsilon = \frac{1}{2}$. For *every* partition $P = \{0 = x_0, \dots, x_n = 1\}$ and every k ,

$$m_k = \inf\{f(x) : x \in [x_{k-1}, x_k]\} = 0$$

because the irrationals are dense in $\mathbb{R}$ (Corollary 1.7), and

$$M_k = \sup\{f(x) : x \in [x_{k-1}, x_k]\} = 1$$

because the rationals are dense (Theorem 1.6). Hence every lower sum is

$$L(f, P) = \sum_{k=1}^n 0 \cdot (x_k - x_{k-1}) = 0$$

and every upper sum is

$$U(f, P) = \sum_{k=1}^n 1 \cdot (x_k - x_{k-1}) = x_n - x_0 = 1.$$

So $U(f, P) - L(f, P) = 1 > \frac{1}{2} = \varepsilon$ for all P , which negates Theorem 17.7. Since that criterion is an “if and only if”, f is not integrable on $[0, 1]$.

Example 18.5 (Thomae's function *is* integrable on $[0, 1]$). Let t be as in Example 9.9. Fix $\varepsilon > 0$ and consider

$$S = \{x \in [0, 1] : t(x) \geq \frac{\varepsilon}{2}\}.$$

S is finite. Since $1/n \rightarrow 0$, only finitely many values $1/n$ exceed $\varepsilon/2$; and for each such n , the preimage $t^{-1}(1/n) \cap [0, 1]$ is contained in the finite set $\{0/n, 1/n, 2/n, \dots, n/n\}$.

Index $S = \{x_0, x_1, \dots, x_N\}$, let $d = \min\{|x_i - x_j| : 0 \leq i \neq j \leq N\}$ be the minimum spacing, and set

$$\delta = \min\left\{\frac{\varepsilon}{6N}, \frac{d}{3}\right\}.$$

Form the partition $P = \{x_i + \delta : 0 \leq i < N\} \cup \{x_i - \delta : 0 < i \leq N\}$ (together with the endpoints), which isolates each point of S inside its own subinterval of width 2δ .

Upper sum. Split the subintervals into those meeting S and those not. On the latter, $M_k < \varepsilon/2$, and their total length is at most 1, contributing less than $\varepsilon/2$. On the former, $M_k \leq 1$ and the total length is at most $N \cdot 2\delta \leq N \cdot \frac{\varepsilon}{3N} = \frac{\varepsilon}{3}$. Hence

$$U(t, P) \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{3} < \varepsilon.$$

Lower sum. The irrationals are dense, so every subinterval contains a point where t vanishes; thus $m_k = 0$ for all k and $L(t, P) = 0$.

Therefore $U(t, P) - L(t, P) < \varepsilon$, and t is integrable with $\int_0^1 t = 0$.

19 Properties of the Integral

19.1 Excusing an endpoint

Theorem 19.1 (Abbott 7.3.2). *If $f : [a, b] \rightarrow \mathbb{R}$ is bounded, and f is integrable on $[c, b]$ for every $c \in (a, b)$, then f is integrable on $[a, b]$. An analogous result holds at the other endpoint.*

Proof. Since f is bounded there is $B > 0$ with $|f(x)| \leq B$ for all $x \in [a, b]$. Fix $\varepsilon > 0$ and choose

$$\delta = \frac{\varepsilon}{4B}.$$

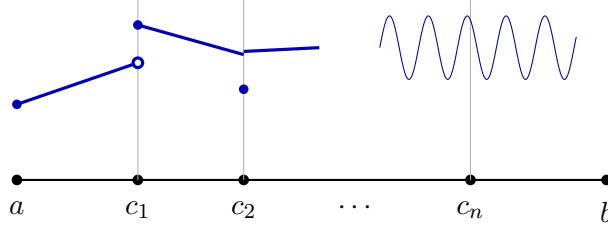
On the sliver $[a, a+\delta]$ the oscillation of f is at most $2B$, so its contribution to $U(f, P) - L(f, P)$ is at most

$$2B \cdot \delta = 2B \cdot \frac{\varepsilon}{4B} = \frac{\varepsilon}{2}.$$

On $[a + \delta, b]$ the function is integrable by hypothesis, so Theorem 17.7 gives a partition Q of $[a + \delta, b]$ with $U(f, Q) - L(f, Q) < \varepsilon/2$. Let $P = \{a\} \cup Q$. Then

$$U(f, P) - L(f, P) = \underbrace{(\text{contribution from } [a, a + \delta])}_{\leq \varepsilon/2} + \underbrace{(\text{contribution from } [a + \delta, b])}_{< \varepsilon/2} < \varepsilon. \quad \blacksquare$$

Corollary 19.2 (Finitely many discontinuities are harmless). *If $f : [a, b] \rightarrow \mathbb{R}$ is bounded and continuous on $[a, b] \setminus \{c_1, c_2, \dots, c_n\}$, then f is integrable on $[a, b]$.*



Theorem 19.3 (Stitching two integrals). *If $f : [a, b] \rightarrow \mathbb{R}$ is bounded and $a < c < b$, then*

$$f \text{ integrable on } [a, b] \iff f \text{ integrable on } [a, c] \text{ and on } [c, b],$$

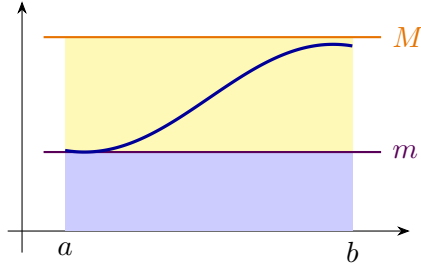
and when these hold,

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

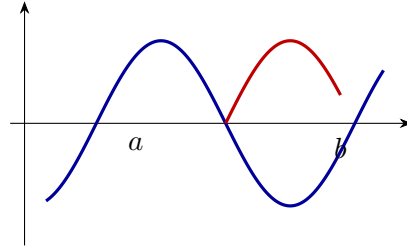
19.2 Algebraic and order properties

Theorem 19.4 (Abbott 7.4.2). *Assume f and g are integrable on $[a, b]$.*

- (i) $f + g$ is integrable, with $\int_a^b (f + g) = \int_a^b f + \int_a^b g$.
- (ii) For $k \in \mathbb{R}$, kf is integrable with $\int_a^b kf = k \int_a^b f$.
- (iii) If $m \leq f(x) \leq M$ on $[a, b]$, then $m(b - a) \leq \int_a^b f \leq M(b - a)$.
- (iv) If $f(x) \leq g(x)$ on $[a, b]$, then $\int_a^b f \leq \int_a^b g$.
- (v) $|f|$ is integrable and $|\int_a^b f| \leq \int_a^b |f|$ (the integral triangle inequality).



$$(iii) \quad m(b - a) \leq \int_a^b f \leq M(b - a)$$



(v) reflecting the negative part gives $|f|$

Proof of (iii) and (iv). (iii) Use the *trivial partition* $P = \{a, b\}$, for which $L(f, P) = m(b - a)$ and $U(f, P) = M(b - a)$. Since $L(f, P) \leq \int_a^b f \leq U(f, P)$ always, the claim follows.

(iv) Let $h = g - f \geq 0$. By (iii) with $m = 0$, $0 = 0 \cdot (b - a) \leq \int_a^b h$. Expanding with (i) and (ii),

$$0 \leq \int_a^b (g(x) - f(x)) = \int_a^b g + \int_a^b (-1)f = \int_a^b g - \int_a^b f,$$

so $\int_a^b f \leq \int_a^b g$. ■

Proposition 19.5 (Scalar multiples, in detail). *If f is integrable on $[a, b]$ and $c \geq 0$, then cf is integrable on $[a, b]$ and $\int_a^b cf = c \int_a^b f$. (The case $c < 0$ requires swapping suprema and infima.)*

Proof. Fix $\varepsilon > 0$. Since f is integrable, Theorem 17.7 gives a partition P of $[a, b]$ with $U(f, P) - L(f, P) < \varepsilon/c$. Write $\widehat{M}_k$ and $\widehat{m}_k$ for the supremum and infimum of f on $[x_{k-1}, x_k]$. Because $c \geq 0$,

$$\sup\{cf(x) : x \in [x_{k-1}, x_k]\} = c\widehat{M}_k, \quad \inf\{cf(x) : x \in [x_{k-1}, x_k]\} = c\widehat{m}_k,$$

so

$$U(cf, P) - L(cf, P) = \sum_{k=1}^n c\widehat{M}_k \Delta x_k - \sum_{k=1}^n c\widehat{m}_k \Delta x_k = c(U(f, P) - L(f, P)) < c \cdot \frac{\varepsilon}{c} = \varepsilon,$$

where $\Delta x_k = x_k - x_{k-1}$. Hence cf is integrable. For the value,

$$L(cf) = \sup_P L(cf, P) = \sup_P cL(f, P) = c \sup_P L(f, P) = cL(f) = c \int_a^b f,$$

and $\int_a^b cf = L(cf)$ because cf is integrable. ■

19.3 Passing limits inside the integral

Theorem 19.6 (Integrable Limit Theorem). *If $f_n \rightarrow f$ uniformly on $[a, b]$ and each f_n is integrable on $[a, b]$, then f is integrable on $[a, b]$ and*

$$\lim_{n \rightarrow \infty} \int_a^b f_n = \int_a^b f.$$

$$\begin{array}{ccc} \left(\int_a^b f_n \right)_{n=1}^\infty & \xrightarrow{\quad} & \int_a^b f \\ \text{IF} \uparrow & & \uparrow \text{THEN} \\ \int_a^b & & \int_a^b \\ \text{IF} \uparrow & \xrightarrow{\text{unif}} & f \end{array}$$

Proof of the limit statement. Fix $\varepsilon > 0$. Using Theorem 19.4(i),(ii) and then (v),

$$\left| \int_a^b f_n - \int_a^b f \right| = \left| \int_a^b (f_n - f) \right| \leq \int_a^b |f_n - f|.$$

Because $f_n \rightarrow f$ uniformly on $[a, b]$, there is $N \in \mathbb{N}$ such that $n \geq N$ implies

$$|f_n(x) - f(x)| < \frac{\varepsilon}{b-a} \quad \text{for all } x \in [a, b].$$

Applying Theorem 19.4(iii) with $M = \varepsilon/(b-a)$, whenever $n \geq N$,

$$\left| \int_a^b f_n - \int_a^b f \right| \leq \int_a^b |f_n - f| \leq \frac{\varepsilon}{b-a} (b-a) = \varepsilon. \quad \text{■}$$

Remark 19.7. Uniformity is essential. There do exist integrable f_n converging *pointwise* to a non-integrable limit — for instance, enumerate $\mathbb{Q} \cap [0, 1] = \{q_1, q_2, \dots\}$ and let f_n be the indicator of $\{q_1, \dots, q_n\}$. Each f_n has finitely many discontinuities, hence is integrable with $\int_0^1 f_n = 0$; but $f_n \rightarrow$ Dirichlet’s function pointwise, which by Example 18.4 is not integrable. This defect is one of the motivations for the Lebesgue integral.

20 The Fundamental Theorem of Calculus

Theorem 20.1 (FTC, Part (i)). *If $f : [a, b] \rightarrow \mathbb{R}$ is integrable and $F : [a, b] \rightarrow \mathbb{R}$ satisfies $F'(x) = f(x)$ for all $x \in [a, b]$, then*

$$\int_a^b f = F(b) - F(a).$$

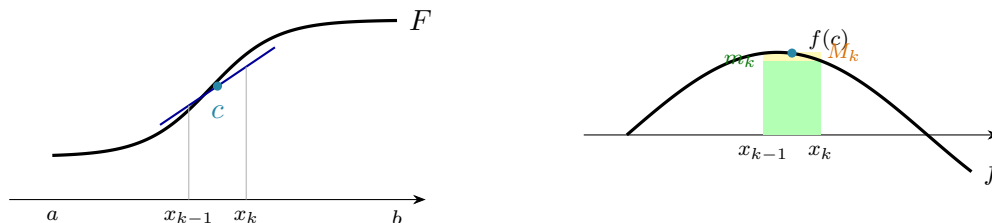
Scratch work. Show that for every partition P ,

$$L(f, P) \leq F(b) - F(a) \leq U(f, P).$$

Then $F(b) - F(a)$ is an upper bound for the set of lower sums and a lower bound for the set of upper sums, so

$$\int_a^b f = L(f) \leq F(b) - F(a) \leq U(f) = \int_a^b f,$$

forcing equality.



Proof. Fix a partition $P = \{a = x_0, x_1, \dots, x_n = b\}$. Since F is differentiable on $[a, b]$ — hence continuous — the Mean Value Theorem (Theorem 12.2) applies on each subinterval: there exists $c_k \in (x_{k-1}, x_k)$ with

$$f(c_k) = F'(c_k) = \frac{F(x_k) - F(x_{k-1})}{x_k - x_{k-1}}.$$

Since $m_k \leq f(c_k) \leq M_k$,

$$m_k \leq \frac{F(x_k) - F(x_{k-1})}{x_k - x_{k-1}} \leq M_k \quad \implies \quad m_k(x_k - x_{k-1}) \leq F(x_k) - F(x_{k-1}) \leq M_k(x_k - x_{k-1}).$$

Summing over k and telescoping the middle,

$$\sum_{k=1}^n m_k(x_k - x_{k-1}) \leq \sum_{k=1}^n \left(F(x_k) - F(x_{k-1}) \right) \leq \sum_{k=1}^n M_k(x_k - x_{k-1}),$$

that is,

$$L(f, P) \leq F(b) - F(a) \leq U(f, P).$$

As P was arbitrary, taking the supremum on the left and the infimum on the right gives $L(f) \leq F(b) - F(a) \leq U(f)$. Since f is integrable, $L(f) = U(f) = \int_a^b f$, so $\int_a^b f = F(b) - F(a)$. ■

Theorem 20.2 (FTC, Part (ii)). *Let $g : [a, b] \rightarrow \mathbb{R}$ be integrable on $[a, b]$ and continuous at $c \in [a, b]$. Define*

$$G(x) = \int_a^x g \quad \text{for } x \in [a, b].$$

Then G is continuous on $[a, b]$, differentiable at c , and $G'(c) = g(c)$.

Remark 20.3. In slogan form: $\frac{d}{dx} \left[\int_a^x g \right] = g(x)$ — differentiating an integral restores the original function. The proof is in Abbott, pp. 235–236.

Remark 20.4 (Integration improves regularity). Rank the conditions from weakest to strongest: integrable, continuous, differentiable, C^∞ . Integration moves a function *up* this ladder, while differentiation moves it down. For example, starting from the Heaviside function g (integrable, discontinuous) and integrating twice:

$$g(x) = \begin{cases} 0 & x < 0 \\ 1 & x \geq 0 \end{cases} \xrightarrow{\int_{-1}^x} G(x) = \begin{cases} 0 & x < 0 \\ x & x \geq 0 \end{cases} \xrightarrow{\int_{-1}^x} \mathcal{G}(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{2}x^2 & x \geq 0 \end{cases}$$

— discontinuous, then continuous, then continuously differentiable.

A second observation: functions that differ on a set of *content zero* integrate to the same value. Thomae's function differs from 0 only on $\mathbb{Q} \cap [a, b]$, and $\int_0^x t = 0$ for every x . Making “content zero” precise is the starting point of measure theory.

Remark 20.5 (An integral form of the remainder). Abbott's Theorem 8.3.1 gives an alternative to Theorem 16.6: if f is $(N+1)$ -times differentiable on $(-R, R)$ with $f^{(N+1)}$ continuous, then

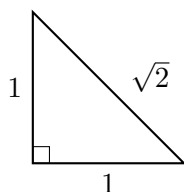
$$f(x) = f(0) + f'(0)x + \underbrace{\frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(N)}(0)}{N!}x^N}_{S_N(x)} + \underbrace{\frac{1}{N!} \int_0^x f^{(N+1)}(t) (x-t)^N dt}_{E_N(x)}.$$

The proof is induction on N using the FTC and integration by parts. The base case $N = 0$ is exactly $\int_0^x f'(t) dt = f(x) - f(0)$; the inductive step takes $u = -f'(t)$, $dv = -dt$, so $v = x - t$.

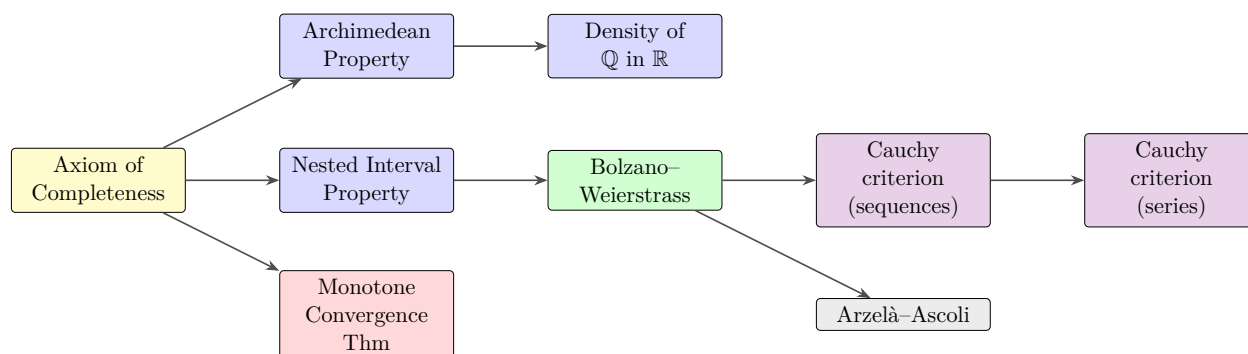
Chapter 6: Constructing $\mathbb{R}$ from $\mathbb{Q}$

21 Where We Have Been

We began by observing that not every geometric length is a ratio of integers: the hypotenuse of the unit right triangle is irrational.



The Axiom of Completeness (Axiom 1.2) codified the intuition that the real line has no holes; it is what let us assert that $\sqrt{2}$ exists, as $\sup\{x \in \mathbb{Q} : x^2 < 2\}$. Its logical consequences stretched across the entire course:



By stating completeness as an *axiom* rather than a theorem, we avoided having to prove it. This chapter asks: could we instead have *constructed* a set with all these properties, starting only from $\mathbb{Q}$?

22 Cauchy Sequences of Rationals

Imagine we have never heard of real numbers. We know $\mathbb{Z}$, we know $\mathbb{Q}$, and we know how to work with

- (i) the arithmetic operations $+$, $-$, $\times$, $\div$;
- (ii) the order $<$ (e.g. $\frac{9}{5} < \frac{28}{15}$);
- (iii) the absolute value $|\cdot| : \mathbb{Q} \rightarrow \mathbb{Q}$ (e.g. $|\frac{9}{5}| = |-\frac{9}{5}| = \frac{9}{5}$).

Note how the rational number $\frac{1}{k}$ takes over the role usually played by ε — we may not use real ε , since reals do not yet exist.

Definition 22.1 (Convergent and Cauchy sequences in $\mathbb{Q}$). A sequence of rational numbers $(q_n)_{n=1}^{\infty}$

- *converges* to $q \in \mathbb{Q}$ if for all $k \in \mathbb{N}$ there exists $N \in \mathbb{N}$ such that $n \geq N$ implies $|q_n - q| < \frac{1}{k}$;
- is *Cauchy* if for all $k \in \mathbb{N}$ there exists $N \in \mathbb{N}$ such that $n, m \geq N$ implies $|q_n - q_m| < \frac{1}{k}$.

Every convergent sequence of rationals is Cauchy (use $|q_n - q_m| \leq |q_n - q| + |q - q_m|$). But — and this is the whole point — *not every Cauchy sequence of rationals converges in $\mathbb{Q}$* .

Example 22.2. The sequence of decimal truncations

$$(q_n) = \left(\frac{1}{1}, \frac{14}{10}, \frac{141}{100}, \frac{1414}{1000}, \frac{14142}{10000}, \dots \right) = (1, 1.4, 1.41, 1.414, 1.4142, \dots)$$

is Cauchy, since $|q_n - q_m| < 10^{-N+1}$ for $n, m \geq N$. Despite our best attempts to forget about real numbers, it looks like it is converging to $\sqrt{2}$. It would be convenient to say that this Cauchy sequence *represents* $\sqrt{2}$.

But representatives are not unique. Newton's algorithm $r_1 = 1$, $r_{n+1} = \frac{1}{2}(r_n + 2/r_n)$ produces

$$(r_n) = \left(1, \frac{3}{2}, \frac{17}{12}, \frac{577}{408}, \frac{665857}{470832}, \dots \right),$$

which also “converges to $\sqrt{2}$ ”. If both (q_n) and (r_n) represent $\sqrt{2}$, we need a way to group equivalent sequences together.

23 Equivalence Classes of Cauchy Sequences

Definition 23.1 (Equivalence relation). An *equivalence relation* on a set $\mathcal{C}$ is a relation $\sim$ among its elements satisfying, for all $a, b, c \in \mathcal{C}$,

- (i) *reflexivity*: $a \sim a$;
- (ii) *symmetry*: $a \sim b$ implies $b \sim a$;
- (iii) *transitivity*: $a \sim b$ and $b \sim c$ imply $a \sim c$.

Definition 23.2 (Equivalence class). The *equivalence class* of $a \in \mathcal{C}$, denoted $[a]$, is the set $\{c \in \mathcal{C} : c \sim a\}$. We call a a *representative* of $[a]$; if $c \sim a$, then c is another representative of the same class.

The equivalence classes generated by an equivalence relation are non-empty, pairwise disjoint, and their union is all of $\mathcal{C}$ — every element lies in exactly one class.

Example 23.3. You have been using equivalence classes for most of your life.

- *The natural numbers.* Take $\mathcal{C} = \{\text{finite collections of objects}\}$ with $A \sim B$ if A and B have the same cardinality. The class containing $\{\text{apple, apple, apple}\}$ is what we call 3. A child holding up two fingers has chosen a representative of the class “2”.

- *The rationals.* $\mathbb{Q}$ consists of equivalence classes of $\{\frac{a}{b} : a, b \in \mathbb{Z}, b \neq 0\}$ under $\frac{a}{b} \sim \frac{c}{d}$ if $ad = bc$. Thus $\frac{3}{2}$ and $\frac{30}{20}$ are the same rational number; we usually pick the reduced ratio as representative.

24 Defining $\mathbb{R}$

Definition 24.1 (The real numbers). Let

$$\mathcal{C} = \{\text{Cauchy sequences of rational numbers}\},$$

and define $\sim$ on $\mathcal{C}$ by

$$(q_n)_{n=1}^{\infty} \sim (r_n)_{n=1}^{\infty} \quad \text{if} \quad (q_n - r_n)_{n=1}^{\infty} \rightarrow 0.$$

The set of *real numbers* $\mathbb{R}$ is the set of equivalence classes of $\mathcal{C}$ under $\sim$.

In words: two Cauchy sequences of rationals represent the same real number when their term-by-term differences converge to zero. Under this definition, the two sequences of Example 22.2 lie in the same class — which is what we *mean* by $\sqrt{2}$.

24.1 Confirming the properties of $\mathbb{R}$

Producing a set is not enough; we must check it behaves as required. One must show that this set of equivalence classes

- (i) contains $\mathbb{Q}$ (in a suitable sense);
- (ii) has well-defined operations $+$, $-$, $\times$, $\div$ satisfying the field axioms;
- (iii) has a well-defined order $<$ compatible with the order on $\mathbb{Q}$; and
- (iv) satisfies the Axiom of Completeness.

Exercise 24.2. Which of properties (i)–(iv) does $\mathbb{Q}$ itself satisfy?

Exercise 24.3. To confirm (i), fix $q \in \mathbb{Q}$. Write down a Cauchy sequence of rationals converging to q , and describe the equivalence class of Cauchy sequences that corresponds to q in this construction.

To give the flavour of the arguments, here is addition.

Definition 24.4. Let $[(q_n)]$ and $[(r_n)]$ be equivalence classes of Cauchy sequences of rationals. Define

$$[(q_n)] + [(r_n)] = [(q_n + r_n)].$$

Read the notation carefully: (q_n) , (r_n) , $(q_n + r_n)$ are particular sequences, while $[(q_n)]$, $[(r_n)]$, $[(q_n + r_n)]$ are equivalence *classes*. To add two classes we choose a representative from each, add them term by term, and take the class of the result. For this to define an operation on classes, the answer must not depend on which representatives we chose.

Proposition 24.5 (Addition is well defined). *If $[(p_n)] = [(q_n)]$ and $[(s_n)] = [(r_n)]$, then $[(p_n + s_n)] = [(q_n + r_n)]$.*

Proof. We must show $(p_n + s_n) \sim (q_n + r_n)$, that is $((p_n + s_n) - (q_n + r_n)) \rightarrow 0$. Fix $k \in \mathbb{N}$.

Since $[(p_n)] = [(q_n)]$ we have $(p_n - q_n) \rightarrow 0$, so there exists $N_1 \in \mathbb{N}$ with $|p_n - q_n| < \frac{1}{2k}$ whenever $n \geq N_1$. Similarly $[(s_n)] = [(r_n)]$ gives $N_2 \in \mathbb{N}$ with $|s_n - r_n| < \frac{1}{2k}$ whenever $n \geq N_2$. Let $N = \max\{N_1, N_2\}$. Then for all $n \geq N$,

$$|(p_n + s_n) - (q_n + r_n) - 0| \leq |p_n - q_n| + |s_n - r_n| < \frac{1}{2k} + \frac{1}{2k} = \frac{1}{k},$$

as desired. ■

Definition 24.6. Multiplication is defined analogously: $[(q_n)] \times [(r_n)] = [(q_n \times r_n)]$.

Exercise 24.7. Show that multiplication is well defined: if $[(p_n)] = [(q_n)]$ and $[(s_n)] = [(r_n)]$, then $[(p_n \times s_n)] = [(q_n \times r_n)]$. *Hint: Cauchy sequences are bounded.*

24.2 Conclusion

The strategy used to construct $\mathbb{R}$ from $\mathbb{Q}$ generalizes: it can be used to “complete” other sets, such as function spaces, so that one can do analysis on them. In outline:

1. Define a *metric* measuring distance between objects in your set. On $\mathbb{Q}$ we used $\text{dist}(q, r) = |q - r|$; on the differentiable functions on $[0, 1]$ one could use $\text{dist}(f, g) = \sup\{|f(x) - g(x)| : x \in [0, 1]\}$.
2. Define Cauchy sequences of elements in terms of that metric.
3. Form equivalence classes of Cauchy sequences.

Remark 24.8 (Further reading). An alternative construction, via *Dedekind cuts*, appears in Abbott pp. 297–302. For a careful treatment of the Cauchy-sequence construction, see Kemp, *Cauchy’s Construction of $\mathbb{R}$* .

References

- [Abb15] Stephen Abbott. *Understanding Analysis*. 2nd ed. Undergraduate Texts in Mathematics. New York, NY: Springer, 2015, pp. XII, 312. ISBN: 978-1-4939-2711-1. DOI: [10.1007/978-1-4939-2712-8](https://doi.org/10.1007/978-1-4939-2712-8). URL: <https://doi.org/10.1007/978-1-4939-2712-8>.